

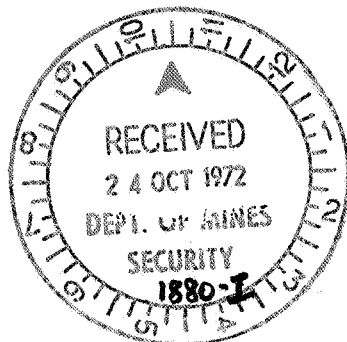
MISCELLANEOUS BORES

APPREETINA BORE

BARATTA No.1

MARLA BORE

~~OT~~
ROBERTSON No. 1.
MALBOOMA COAL BORES 1, 2, 3, 5



346

LATITUDE LONGITUDE

BORE HOLE RECORD

INDEX NO.

27° 45' S

131° 15' E

NAME OF BORE/WELL: APPRETTINA BORE

GEOLOGIC AREA : BETWEEN OFFICER / PEDIRKA BASIN

_____ :

_____ :

_____ :

NOTES ON LOCATION :DATE COMPLETED :ABBREVIATED LOG :REFERENCES : located on Wintinna Topog. 1:250,000 P.D.
U.S.S. A. QuarterREMARKS :

LATITUDE LONGITUDE

BORE HOLE RECORD

INDEX NO.

31° 54' S

139° 28' E

NAME OF BORE/WELL: BARATTA NO.1..

GEOLOGIC AREA : ADELAIDE GEOSYNCLINE

_____ :

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NOTES ON LOCATION : 60 miles NW YuntaDATE COMPLETED : 1967ABBREVIATED LOG : see above.

REFERENCES : Baratta GML 147 Cominco : D.M. ENV. 120/67
 Can. Zinc File 1958-60 : D.M. ENV. 57 (P.G. Seale, NSHC)
 Cominco GML 102 : D.M. ENV. 31/66 (L.G. Nixon)

REMARKS : Hole Soffered 55° on bearing of 332° magnetic.
 Purpose of Drilling to investigate possible
 galena mineralization at depth in vicinity of
 old Cross Canada and Rutaby workings and to
 assess for Cu in unexposed Tindellina shale [elite correlated with (P. anomalies)]

approx. log only. (see D.M. ENV. 120/67 for X-section)

0-25 river gravels + alluvium

25-65 carbonaceous wavy bedded siltst. & c/s

65-130 brown sandst., soft, massive; narrow beds
carbonaceous phyllite130-150 brown g/fite; some narrow carbon.
clay bands.150-180 indurated brown sandstones, MnO₂180-210 soft sandstone with mudstone bands
and hard ferrug. bands carrying MnO₂

210-250 carbonaceous siltstone (as for 25-65)

250-280 as for 65-130

280-295 as for 150-180

295-425 as for 65-130

425-450 calc. carb. sandstone

450-523 siltstone & sandstone

fresh pyrite first observed at 300'

Pyrite content from 300'-523' T.D. varies
from 41 to 3% volume.

LATITUDE LONGITUDE

BORE HOLE RECORD

INDEX NO.

27° 14'

133° 36'

NAME OF BORE/WELL: MARLA BORE

GEOLOGIC AREA : BETWEEN OFFICER / PEDIRKA BASINS

_____ :

_____ :

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NOTES ON LOCATION :DATE COMPLETED :ABBREVIATED LOG : 0-Q-94 - ? O.R. - 339 T.D.REFERENCES : DFM Stat. Bore 347REMARKS :

BORE HOLE ASSAYS

352
INDEX NO.

[illegible]

ASSAY TECHNIQUE: A/A: MODEL TOR NO. 2373/72

LATITUDE LONGITUDE
APPROX. ONLY

BORE HOLE RECORD

INDEX NO.

23° 40'S

131° 11'E

NAME OF BORE/WELL: MALBOOMA COAL BORES 1,2,3,5

GEOLOGIC AREA : GAWLER CRATON

_____ :

_____ :

_____ :

NOTES ON LOCATION : In an area approx. 22.5 miles west of Tancoola ; Mulgathing Station.

DATE COMPLETED : 1959-61

ABBREVIATED LOG :

REFERENCES : Mining Rev. 116 p45.
D ↑ M 779/59 : 21/2/58.
(E.R. Hillwood)

REMARKS :

Hole positions not given in MR116.

HOLE 1 0-27 Trav. 1/5 + clay
27-59 clay, limonitic, sandy layers
59-69 black and brown carbon.
clay 59' & 69' with silty
and mic. clay bands between
69-155 carbon. clay & lignite
155-193 sand & decomp. gneiss
193-225 granite gneiss

HOLE 2 0-25 sand, 1/5, sandy clay
25-58 weath. granite

HOLE 3 0-99 sand, calc. sandy clay,
gritty towards base
99-153 weath. granite

HOLE 4 0-77 clay, siltstone
77-135 weath. granite

HOLE 5 0-76 sand, clay
76-123.5 weath. granite

BORE HOLE ASSAYS

INDEX NO.

			NAME OF BORE/WELL: MALBOOMA NO.1												
	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Ag	Mo	U				
Assay No.	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.				
25765	30	60	CUTTINGS	See M.B. 116.		5	10	15	<1	<3	-				
25766	60	80				10	10	230	<1	<3	-				
25767	80	100				5	<5	10	<1	<3	-				
25768	100	120				70	25	85	<1	<3	3				
25769	120	140				45	25	35	<1	<3	<2				
25770	140	160				25	25	45	<1	<3	-				
25771	160	180				30	25	120	<1	<3	-				
25772	180	200				150	25	75	<1	<3	-				
25773	200	222				110	20	110	<1	<3	8				
ASSAY TECHNIQUE: A/P AMDEL JOB NO. 276/72															

- : not assayed.

BORE HOLE ASSAYS

INDEX NO.

			NAME OF BORE/WELL: MALBOOMA NO. 2.											
	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Ag	Mo				
Assay No.	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.				
25774	20	40	CUTTINGS	See m.p. 16		15	15	25	<1	<3				
25775	40	52	"			15	10	40	<1	<3				

ASSAY TECHNIQUE: A/AAMDEL JOB NO. 2761/72

[illegible]

BORE HOLE ASSAYS

359
INDEX NO.

			NAME OF BORE/WELL: MALBOOMA NO. 5													
	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Ag	Mo	U					
Assay No.	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.					
25776	80	100	CUTTINGS	See M.R. 116		50	25	30	<1	3						
25777	100	123	"			15	20	65	<1	3	4					
ASSAY TECHNIQUE: A/PAMDEL JOB NO. 2761/72																

STUART SHELF

E. & W. S. BEDA BORE

D.M. BUTE D.D.H. No.5

CLARENCE RIVER BASIN OIL WOOMERA No.1

D.M. ROOPENA STRATIGRAPHIC D.D.H. No.1

With reference to the Pernatty Lagoon
(Mt. Gunson Mine) Cu - Mn - Ba mineralization
and the Pandurra Copper Prospect.



BORE HOLE ASSAYS

			NAME OF BORE/WELL: BEDA BORE												
	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Ag						
Assay No.	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.						
101	?	?	fragments			15	20	230	<1						
102	"	"	"			20	20	230	<1						
103	"	"	"			15	20	230	<1						
104	"	"	"			15	20	230	<1						
105	"	"	"			25	20	190	<1						
106	"	"	"			20	20	190	<1						
107	"	"	"			25	20	180	<1						
108	"	"	"			20	20	180	<1						
109	"	"	"			25	20	180	<1						
110	"	"	"			20	20	190	<1						
111	"	"	"			25	20	200	<1						
112	"	"	"			30	25	180	<1						
ASSAY TECHNIQUE: AMDEL JOB NO. 2373/72															

BORE HOLE ASSAYS

INDEX NO.

			NAME OF BORE/WELL: BEDA BORE												
	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Ag.						
Assay No.	From	To				pp.m.	pp.m.	pp.m.	pp.m.						
113	?	?	fragments			25	25	160	<1						
114	"	"	"			25	20	160	<1						
115	"	"	"			25	25	220	<1						
116	"	"	"			20	25	170	<1						
117	"	"	"			20	25	160	<1						
118	"	"	"			25	25	160	<1						
119	"	"	"			30	25	170	<1						
120	"	"	"			30	25	170	<1						

ASSAY TECHNIQUE: AMDEL JOB NO. 2373/72

FORM 6 JOB

953/72

TT	Analytical Sample No.	Petrological Sample	ppm Cu	depth in feet	Pb	Zn	Petrography
1	A 991/11	P925/71	10	120'	10	100	altered igneous rock
2	80	... 6 ...	15	130'	10	55	" " "
3	81	... 7 ...	(120)	150'	10	35	metasiltstone
4	82 X	... 8 ...	30	200'	15	85	" " "
5	83	... 9 ...	(710)	300'	15	85	altered porphyritic volcanic (andesite?)
6	84	P830/71	10	400'	15	150	chloritized igneous rock (volcanic?)
7	85	... 1 ...	10	500'	5	35	metasiltstone
8	86	2	(150)	525' 2"	10	35	metasiltstone / contains altered igneous
9	87	3	95	550'	(20)	120	altered dolomite or amphibolite
10	88	4	10	600'	15	140	chloritized amphibolite
11	STD 52						
12	89	5	65	657'	(20)	140	chloritized amygdaloidal basalt
13	90	6	(230)	672'	(45)	130	altered basic volcanic
14	A 991/71	P937/71	15	693'	10	100	metasiltstone
15	82 X						
16	BLANK		—		—	—	
17							Code C1
18							
19							Results in ppm
20							

Bute. Stratigraphic D.D. Hole. No. 5
 Summary of Log. - B.P. Thomson, SA. Dept.

from to
 0' - 69'
 69' - 118'
 118' - 133'
 133' - 525' 2"
 525' 2" - 693'
 693' - 701'

Sand + clay - (Quaternary)
 Weathered Dolomite - (Cambrian)
 Altered basic igneous rock. (Precambrian)
 Metasiltstone and altered basic volcanics,
 Amphibolite, ^{basic volcanics} and metasiltstone
 Metasiltstone

END OF HOLE

1119.
 1190.

Intersected by Aserco
 panned by Aserco

LATITUDE LONGITUDE

BORE HOLE RECORD

INDEX NO.

NAME OF BORE/WELL: BUTE D.D.H. 5

GEOLOGIC AREA : YORKE PENINSULA

_____ :

_____ :

_____ :

NOTES ON LOCATION :DATE COMPLETED : 1971ABBREVIATED LOG :

0-69 sand & clay (Quaternary)

69-118 weathered siltstone (Cambrian)

118-133 altered basic igneous rock (Precambrian)

133-525.2 metasilstone and altered basic volcanics.

525.2-693 amphibolite, basic volcanics and metasilstone.

693-701 metasilstone.

END OF HOLE

REFERENCES :

Verba & B. Thompson DTM.

REMARKS :

Provisional log by B. Thompson, S.A. DTM.

BORE HOLE ASSAYS
(By courtesy SA 50 PM - refer B. Thompson)

		NAME OF BORE/WELL: BUTE NO. 5										
Assay No.	Footage		Sample Type	Lithology	Formation Petrology sample.	Cu	Pb	Zn				
	From	To				p.p.m.	p.p.m.	p.p.m.				
A 979/71		120	CORR	alt. ign. rock.	P 925/71	10	10	100				
A 980/71		130		alt. ign. rock.	P 926/71	15	10	55				
A 981/71		150		metasiltstone	P 927/71	120	10	35				
A 982/71		200		metasiltstone	P 928/71	30	15	85				
A 983/71		300		alt. porph. volc. (andesite?)	P 929/71	710	15	85				
A 984/71		400		chlor. ign. rock (volcanic?)	P 930/71	10	15	150				
A 985/71		500		metasiltstone	P 931/71	10	5	35				
A 986/71		525.2		metasiltstone/ alt. ign. contact	P 932/71	150	10	35				
A 987/71		550		alt. dolerite or amphibolite	P 933/71	95	20	120				
A 988/71		600		chloritised amphibolite	P 934/71	10	15	140				
A 989/71		657		alt. basic volc.	P 935/71	65	20	140				
A 990/71		672		alt. basic volc.	P 936/71	230	45	130				

ASSAY TECHNIQUE: A/AMDEL JOB NO. 953/72

A 991/71	693	metasiltstone	P 937/71	15	10	100
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BORE HOLE ASSAYS

INDEX NO.

		NAME OF BORE/WELL: BUTE NO. 5												
Assay No.	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn			
	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.			
26201	69	74.5	CORR	weath. dol. (sulph.?)	Cambrian	15	10	75	10	10	3600			
26202	80.2	90		"		420	70	200	20	25	1.8%			
26203	99.5	101		"		1500	260	420	85	60	1700			
26204	106	115		"		110	<5	30	<5	30	140			
26205	115	118		basal conglom. of dol.		10	<5	35	<5	40 30	340			
26206	118	125.8		alt. basic ign. rock	Precambrian	10	<5	120	20	70	1400			
26207	125.8	134		"		35	<5	100	20	70	1300			
26208	134	140		metasiltstone		140	10	55	20	50	820			
26209	140	150		"		95	5	50	10	35	520			
26210	150	160		"		50	5	35	15	40	200			
26211	160	170		" ?		20	10	45	20	40	260			
26212	170	180		"		90	10	60	20	40	570			

ASSAY TECHNIQUE: A/P AMDEL JOB NO. 3107/72 Batch 12

N.B. 26212 contaminated with Zn from sampling device.

BORE HOLE ASSAYS

INDEX INC.

			NAME OF BORE/WELL: BUTE NO. 5											
Assay No.	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Min			
	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.			
26213	180	190	CORE	metasiltstone?	Pre cambrian	85	15	45	15	40	590			
26214	190	200		metasiltst.		85	10	55	15	40	410			
26215	200	210		"		50	5	75	10	30	410			
26216	210	220		"		220	5	45	10	30	420			
26217	220	230		"		80	5	45	5	30	380			
26218	230	240		"		30	5	65	5	30	400			
26219	240	250		"		80	5	70	5	30	430			
26220	250	260		"		50	10	120	5	40	410			
26221	260	270		"		75	5	90	10	30	420			
26222	270	280		basic volc.		40	5	120	25	65	850			
26223	280	290		"		20	5	120	35	60	680			
26224	290	300		"		140	5	100	35	50	1400			

ASSAY TECHNIQUE: A/AANDEL JOB NO. 310717 Batch 12.

BORE HOLE ASSAYS

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INDEX INC.

			NAME OF BORE/WELL: BUTTE NO. 5											
Assay No.	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn			
	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.			
26225	300	310	COAR	Basic volc.	Precamb.	150	5	100	20	50	1600			
26226	310	320				2200	10	80	30	40	1900			
26227	320	330				2900	5	65	30	30	5100			
26228	330	340				65	5	75	30	55	590			
26229	340	350				15	5	65	20	50	520			
26230	350	360				10	5	65	25	50	1110			
26231	360	370				15	15	110	25	50	1160			
26232	370	380				35	10	95	25	50	1180			
26233	380	390				100	10	100	40	50	1700			
26234	390	400				120	5	120	50	85	1600			
26235	400	410				120	10	100	50	55	1800			
26236	410	420				100	10	120	60	75	11400			

ASSAY TECHNIQUE: AAANDEL JOB NO. 26225:- 3107/72 Batch 12
26226 - 26236 :- Batch 13, 3107/72.

BORE HOLE ASSAYS

377
INDEX INC.

		NAME OF BORE/WELL: BUTTE NO. 5												
Assay No.	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn			
	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.			
26237	420	430	CORE	Basic volc.	Precambrian	55	10	120	55	80	2400			
26238	430	440		metagabbro?		180	10	85	110	55	740			
26239	440	450		"		70	10	70	25	35	1600			
26240	450	460		"		40	10	80	15	35	1170			
26241	460	470		"		55	5	35	15	30	720			
26242	470	480		"		55	5	35	15	25	910			
26243	480	490		"		130	30	35	15	30	1600			
26244	490	500		"		60	5	15	10	20	340			
26245	500	510		"		50	5	30	10	35	260			
26246	510	520		"		220	10	55	10	35	1110			
26247	520	530		"		100	10	50	20	40	930			
26248	530	540		Basic volc.		600	10	80	25	60	770			

ASSAY TECHNIQUE: A/A AMDEL JOB NO. 3107/77 Batch 13, 14.

BORE HOLE ASSAYS

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INDEX INC.

			NAME OF BORE/WELL: BUTE NO. 5											
	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn			
Assay No.	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.			
26249	540	550	CORE	basic volc.	Precamb.	65	10	120	35	70	670			
26250	550	560		basic volc. ± chertophy.		130	10	120	35	80	1500			
26251	560	570		basic volc. tr. cpy.		80	5	110	30	70	720			
26252	570	580		basic volc.		120	70	140	35	80	1300			
26253	580	590		"		90	10	150	35	90	890			
26254	590	600		basic volc. tr. cpy.		250	5	100	35	85	1300			
26255	600	610		" tr. cpy.		510	5	120	35	85	930			
26256	610	620		metasiltstone		130	10	140	55	55	900			
26257	620	630		basic volc.		10	15	160	50	35	1200			
26258	630	640		"		15	10	180	50	35	930			
26259	640	650		"		95	20	200	60	110	960			
26260	650	660		"		90	20	180	55	110	1800			

ASSAY TECHNIQUE: A/A AMDEL JOB NO. 3107/72 BATCH 14.

BORE HOLE ASSAYS

			NAME OF BORE/WELL: BUTE NO. 5												
	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn				
Assay No.	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.				
26261	660	670	CORE	Basic volc. + metasilicst.	Precamb.	390	20	100	55	40	720				
26262	670	680				"	70	10	120	55	50	500			
26263	680	690				"	50	10	120	55	40	440			
26264	690	700					35	10	110	50	30	390			

TITUDE LONGITUDE

BORE HOLE RECORD

INDEX NO.

NAME OF BORE/WELL: WOOMERA BORE

GEOLOGIC AREA : STUART SHELF

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NOTES ON LOCATION : NW of Mt. Gunson

DATE COMPLETED : 1958, Clarence River Basin Oil.

ABBREVIATED LOG : 0-12' Recent soil ; 12'-263' Corraberra Sandstone equivalent - red sandy siltstone and f.g. red sandstone.
 (after Thompson 1968) 263'-785' Woomera shale - red and green laminated siltstone and shale. 785'-1119' Whyalla Sandstone - red, pink

REFERENCES :

and cream f.g.-m.g. feldspathic sandstone. || DISCONFORMITY || 1119' - 1490' Woomera Dolomite - dark grey laminated
 dolomite and siltstone || DISCONFORMITY || Pandurra formation
 1490' - 2006'.

Thompson B.P. & Johnson J.E., 1968. Marinoan Stratigraphy, Port Augusta Region: Quant. geol. notes 286. S.S.A.

REMARKS :

BORE HOLE ASSAYS

381
INDEX NO.

			NAME OF BORE/WELL: WOOMERA BORE											
say No.	Footage		Sample Type	Lithology as apparent during sampling	Formation	Cu	Pb	Zn	Co	Ni	Mn			
	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.			
26001	10	23	CORE	shale/siltstone	Unnamed member -	25	15	100	20	35	970			
26002	23	30		shale	Corraberra Sandstone member of	25	15	110	20	35	1000			
26003	30	36		shale	Tent Hill Formation.	25	10	90	20	30	1300			
26004	36	46		shale		50	15	100	20	35	1000			
26005	46	56		shale		30	10	90	20	35	1200			
26006	56	66		shale	Red sandy siltstone and f.g.	50	10	90	20	35	1200			
26007	66	76		shale	red sandstone, sedimentary	30	15	90	20	35	1200			
26008	76	86		shale	flump structures	5	15	90	20	35	1200			
26009	SAMPLE NUMBER NOT USED				Small scale crossbedding.									
26010	86	90		shale		5	10	75	15	30	680			
26011	90	100		shale		5	< 10	35	10	15	690			
26012	100	110		shale		30	10	100	20	35	1000			

AY TECHNIQUE: A/PADEL JOB NO. 3107/72
Batch 1

BORE HOLE ASSAYS

382
INDEX INC.

NAME OF BORE/WELL: Woomera Bore													
Core Sample No.	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn		
	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.		
6013	110	120	CORE	shale	Corraberri	25	10	250	20	30	1100		
6014	120	130		shale	Sandstone	75	10	320	20	35	1300		
6015	130	140		shale	Equivalent.	85	10	175	20	35	1100		
6016	140	150		shale/conglom.		20	10	90	10	15	890		
26017	150	160		shale/arkose		50	10	170	20	35	1500		
26018	160	170		sand/conglom.		10	5	80	15	30	800		
26019	170	180		sandy conglom.		5	5	80	10	20	700		
26020	180	190		shale/sandstone		5	5	95	20	30	2200		
26021	190	200		shale-sandst.		5	10	100	15	20	1000		
26022	200	210		shale		5	10	180	10	15	100		
26023	210	220		fine-sandst./siltstone		5	10	80	5	10	45		
26024	220	240		shale		5	10	110	15	25	110		

ANALYTICAL TECHNIQUE: A/ANDEL JOB NO. 3107/72 Batch 2.

BORE HOLE ASSAYS

INDEX NO.

NAME OF BORE/WELL: WOOMERA BORE												
ay No.	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn	
	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	
6025	230	240	CORE	shale/cundst.	Corraberra	5	10	130	20	30	260	
6026	240	250		siltstone/shale	Sandstone Equivalent.	5	15	110	20	35	180	
6027	250	260		" "		5	10	100	20	35	180	
26028	260	270		" "	263'	5	10 15	80 110	20	35	280	
26029	270	280		" "	Woomera Shale.	35	5	75	20	35	290	
26030	280	290		" "		40	5	80	20	30	380	
26031	290	300		shale		5	10	90	20	35	300	
26032	300	310		"		5	10	110	20	35	310	
26033	310	320		"		10	10	90	20	35	320	
26034	320	330		"		5	10	75	20	40	300	
26035	330	340		"		10	5	90	20	35	340	
26036	340	350		"		100	10	95	20	35	320	

Y TECHNIQUE: A/A AMDEL JOB NO. 3107/72 Batd 2.

BORE HOLE ASSAYS

INDEX NO.

			NAME OF BORE/WELL: WOOMERA BORE											
say No.	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn			
	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.			
6037	350	360	CORE	shale	Woomera Shale	5	10	95	20	35	320			
26038	360	370		"	Red green laminated siltstone and shale.	10	10	90	20	35	320			
26039	370	380		"		25	10	95	20	35	330			
26040	380	390		"		30	10	95	20	35	360			
26041	390	400		"		10	10	120	15	35	360			
26042	400	410		"		5	15	140	15	35	330			
26043	410	420		"		5	15	90	20	40	340			
26044	420	430		"		10	10	90	20	40	340			
26045	430	440		"		5	10	100	20	35	340			
26046	440	450		"		40	10	95	20	40	370			
26047	450	460		"		50	10	130	20	35	380			
26048	460	470		shale turbidites		40	10	95	20	40	380			

AY TECHNIQUE: A/ A AMDEL JOB NO. 3107/72 Batch 3.

BORE HOLE ASSAYS

INDEX NO.

			NAME OF BORE/WELL: WOOMERA BORE											
say No.	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn			
	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.			
26061	590	600	CORE	shale	WOOMERA SHALE	5	10	100	20	40	390			
26062	600	610		"		5	10	95	20	40	340			
26063	610	620		"		5	5	95	15	40	390			
26064	620	630		"		15	5	90	20	40	350			
26065	630	640		"		50	5	95	20	35	720			
26066	640	650		"		25	5	90	20	40	480			
26067	650	660		"		10	10	95	20	50	340			
26068	660	670		"		5	10	95	20	50	350			
26069	670	680		"		5	10	90	15	40	430			
26070	680	690		"		5	10	95	20	40	380			
26071	690	700		"		5	15	100	20	40	390			
26072	700	710		"		5	15	100	15	40	440			

AY TECHNIQUE: AARDEL JCB NO. 3107/72 B.A.D. 445

38.
INDEX NO.

Footage			Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn				
Assay No.	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.				
26073	710	720	CORE	shale	Woomera SHALE	5	15	100	20	40	440				
26074	720	730				5	15	110	20	35	450				
26075	730	740				5	25	130	20	40	510				
26076	740	750				5	25	140	20	40	740				
26077	750	760				5	25	130	20	40	670				
26078	760	770				5	25	120	15	35	1300				
26079	770	784				5	25	110	15	40	1700				
26080	784	791		feldspathic sandstone	Parnatty Grit	5	15	95	<5	10	660				
26081	791	800		gritty sandstone		5	<5	30	<5	5	1300				
26082	800	810				5	<5	70	<5	<5	740				
26083	810	820		grit		5	<5	45	<5	<5	740				
26084	820	830				5	<5	60	<5	<5	750				

ASSAY TECHNIQUE: A/A ANDEL JOB NO. 3107/72 Batch 445

BORE HOLE ASSAYS

INDEX NO.

			NAME OF BORE/ WELL: WOOMERA BORE											
Assay No.	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn			
	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.			
26097	970	980	CORE	grit	PERNATTY GRIT	10	5	120	<5	5	630			
26098	980	1000				10	<5	50	<5	5	540			
26099	1000	1040				15	10	130	<5	5	560			
26100	1040	1060				5	<5	60	<5	<5	910			
26101	1060	1070				5	5	75	<5	10	1700			
26102	1070	1080				5	5	110	<5	10	840			
26103	1080	1090				5	5	130	<5	15	490			
26104	1090	1100				5	45	70	<5	10	910			
26105	1100	1105				200	100	320	<5	10	1400			
26106	1105	1110				10	80	95	<5	10	2300			
26107	1110	1115						Lost SAMPLE						
26108	1115	1119				25	10	90	<5	30	4600			

ASSAY TECHNIQUE: A/A ANDERSON JOB NO. 3107/72 Batch 6

BORE HOLE ASSAYS

INDEX NO.

			NAME OF BORE/WELL: WOOMERA BORE												
	Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn				
Assay No.	From	To				b.p.m.	b.p.m.	b.p.m.	b.p.m.	b.p.m.	b.p.m.				
26109	1119	1125	CORE	1119' DISCONTINUITY grey shale shale <											

BORE HOLE ASSAYS

INDEX NO.

			NAME OF BORE/WELL: BOOMERA BORE												
Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn					
Assay No.	From				To	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.				
26133	1360	1370	CORE	shale	Woolalla Dolomite Equiv.	30	50	160	15	40	680				
26134	1370	1380				30	50	75	15	30	720				
26135	1380	1390				30	110	110	15	30	760				
26136	1390	1410				25	20	120	5	30	1500				
26137	1410	1430				25	20	80	15	30	1300				
26138	1430	1450				30	50	190	15	30	1500				
26139	1450	1460				110	100	65	15	110	2400				
26140	1460	1470				35	90	230	15	30	3100				
26141	1470	1480				30	80	170	10	30	4000				
26142	1480	1485				30	160	100	15	25	5800				
26143	1485	1490		35	150	390	15	20	170						
26144	1490	1495		1490 DISCONFORMITY gltite-grit Fennelly Grit		80	30	220	<5	20	9300				
ASSAY TECHNIQUE: AA ANDEL JOB NO. 3107/72. PAGE 8.															



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PLEASE ADDRESS MAIL TO:
ASARCO (AUSTRALIA) PTY. LTD.
P.O. BOX 6010
HAY STREET EAST, PERTH 6000
WESTERN AUSTRALIA

October 8, 1974.

The Director of Mines,
Department of Mines of South Australia,
Box 38, Rundle Street P.O.,
SOUTH AUSTRALIA, 5000.

Dear Sir,

YOUR REFERENCE DM.131/71 BPT:PAL

In reply to your letter of the 27th September, the
missing figures from Mr. Warne's report on the Woomera No. 1
bore are as follows:

<u>Sample No.</u>	<u>Interval</u>		<u>Cu</u>	<u>Pb</u>	<u>Zn</u>	<u>Co</u>	<u>Ni</u>	<u>Mn</u>
	<u>Feet</u>							
A26157	1600	-1610	<5	5	55	<5	5	85
58	1610	-1620	<5	<5	45	<5	5	75
59	1620	-1632	5	5	45	<5	5	75
60	1632	-1642	<5	5	65	<5	5	75
61	1642	-1654	<5	5	55	<5	10	95
62	1654	-1665	<5	5	55	<5	10	95
63	1665	-1677.5	<5	<5	30	<5	5	100
64	1677.5	-1688	5	<5	25	<5	5	110
65	1688	-1700	10	<5	30	<5	10	100
66	1700	-1720	5	<5	50	<5	10	110
67	1720	-1732	5	<5	40	<5	10	130
68	1732	-1744	5	<5	30	<5	20	130

All values are in parts per million.

If any further omissions are found, please do not
hesitate to bring them to my attention so that the figures
can be supplied if possible.

Yours faithfully,

Michael Kriewaldt

M. Kriewaldt

NOTED
ref.

BORE HOLE ASSAYS

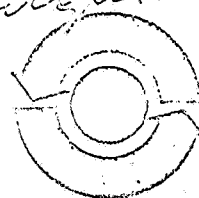
INDEX NO.

Footage			Sample Type	Lithology	Formation	Cu	Pb	Zn	Co	Ni	Mn				
Assay No.	From	To				p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.	p.p.m.				
26181	1379	1391	CORE	felspathic sandstone rare thin siltstone members.	Pemalty Grit.	<5	<5	115	<5	20	1140				
26182	1391	1903				<5	<5	30	<5	20	1140				
26183	1903	1915				<5	<5	95	<5	20	120				
26184	1915	1927				5	<5	95	<5	10	130				
26185	1927	1939				5	110	95	<5	10	270				
26186	1939	1954				5	<5	95	<5	20	130				
26187	1954	1964				5	5	120	<5	20	110				
26188	1964	1976				5	<5	65	<5	10	110				
26189	1976	1988				5	<5	55	<5	10	100				
26190	1988	1998.5				5	<5	55	<5	10	130				
26191	1998.8	2005				5	<5	50	<5	5	160				
					END OF HOLE										

ASSAY TECHNIQUE: AA ANAL JOB NO. 3107/77 BOSTON II.

THE AUSTRALIAN MINERAL DEVELOPMENT CORPORATION

Handwritten signature
P. A. Young



PLEASE ADDRESS ALL CORRESPONDENCE TO THE DIRECTOR.

OUR REFERENCE:

MP1/13/0

YOUR REFERENCE:

11th December, 1967

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The Director,
S.A. Department of Mines,
Box 38, Rundle Street P.O.,
ADELAIDE, 5001.

REPORT MP1152-68

YOUR REFERENCE:

Application dated 11/10/67

MATERIAL:

Drill Core Samples (8)

LOCALITY:

Roopena Diamond Drill Hole No.1
(Footages as listed)

IDENTIFICATION:

P502/67 - P509/67

DATE RECEIVED:

11/10/67

WORK REQUIRED:

Petrographic descriptions.

Investigation and Report by: A.R. Turner

Acting Officer in Charge, Mineralogy Section: N.A. Trueman

P.A. Young
Director.

INVESTIGATION OF THE ROOPENA VOLCANICS

PART 7

1. INTRODUCTION

The samples described in this report were obtained from Roopena Diamond Drill Hole No.1 between the 428 feet 6" and 490 feet 5" level. One sample of rock chips from Beda Bore believed to belong to the Roopena Volcanics is described.

Those carbonates described as calcite have been identified by microchemical staining techniques. Those described as carbonates did not stain and thus are some other carbonate than calcite.

2. PETROGRAPHY

Sample P502/67 (Top) TS19938

Location:

428 feet 6 inches - 428 feet 7½ inches.

Rock Name:

Completely replaced ?volcanic rock.

Hand Specimen:

A pale fawn, massive and intensely chloritized volcanic rock. This rock grades rather abruptly into P502 described from the base of this section of the core. The contact between the two rocks is at approximately 45° to the axis of the core.

Thin Section:

A visual estimate of the constituents gives the following:

	<u>%</u>
Carbonate	35
Epidote	20
Chlorite	25
Quartz	10
Clay minerals	10
Andesine	Accessory
Opagues	Accessory
Sphene	Accessory

This sample is completely altered and displays no relict textures that could provide an indication as to the nature

2.

of the primary rock type. The rock is composed predominantly of masses of crystalline carbonate, only traces of which are calcite, set in a mass of chlorite, epidote and secondary silica. Clay minerals are also abundant. The only relict mineral that has survived this period of alteration is a trace of the plagioclase andesine. One crystal shape that can be interpreted as a completely pseudomorphed phenocryst of feldspar was found. This crystal is now composed of a rim of carbonate enclosing a core of microcrystalline clay minerals.

Special Features:

The presence of former feldspar phenocrysts and relict andesine would suggest that this rock is a severely altered porphyritic andesite. The alteration may be related to ground-water solutions percolating along the contact with the sediments at the base of the volcanic sequence.

History:

Igneous ?extrusive. Extensively altered.

Sample P502/67: (Bottom) TS19989

Location:

428 feet 7½ inches - 428 feet 9 inches.

Rock Name:

An altered, porphyritic, amygdaloidal ?andesite.

Hand Specimen:

A fine-grained, massive, bleached, strongly amygdaloidal volcanic rock. The rock has a pale fawn colouration and the amygdules are deep green. Some minor fractures traverse the sample.

Thin Section:

This rock compositionally resembles the previous sample, however, relict textural features are better preserved and allow an interpretation of the nature of the primary rock. Amygdules are common and occupy some 30% of the sample. These are generally formed of a rim of carbonate enclosing a core of fine-grained, pale-green chlorite. Clay minerals and minor silica may occur in smaller amounts. The host is almost completely altered but can be seen to have been composed of a felted mass of subradiating plagioclase laths that are now completely replaced by carbonate and chlorite. Former phenocrysts of feldspar have a random distribution.

Special Features:

This rock is probably closely related to the previously described sample. The alteration is similar in nature and origin.

History:

Igneous extrusive. Severely altered.

Sample P503/67 (Top): TS19990

Location:

430 feet 1 inch - 430 feet 2 inches.

Rock Name:

Altered, amygdaloidal, porphyritic ?andesite.

Hand Specimen:

A fine-grained, amygdaloidal, mottled green rock. The sample is in contact with a darker clastic rock. The contact is at an angle of approximately 30° to the core axis.

Thin Section:

A visual estimate of the constituents gives the following:

	<u>%</u>
Clay minerals	30
Carbonate	20
Chlorite	30
Epidote	3
Hydrated-iron oxides	15
Opagues	2

The rock is composed of an altered mass of subradiating plagioclase laths that have been variously replaced by carbonate and/or clay minerals. The interstitial areas are occupied by chlorite, carbonate, disseminated opaques and minor epidote. There is a tendency for the grain-size to coarsen as the contact with a clastic sandstone is approached, and hydrated iron-oxides and opaques increase in abundance towards the contact. Amygdules are quite common and are usually formed of a rim of chlorite enclosing a core of carbonate. Occasionally a fibrous, radiating clay mineral occupies the amygdules.

At a distance of approximately 10 mm from the contact a structure becomes evident. Here there is an alignment of the plagioclase laths which parallels the plane of the contact. This structure becomes more marked as the contact is approached. Immediately at the contact the structure disappears and the rock assumes the habit of an altered devitrified glass. The contact is quite sharp.

History:

Igneous extrusive and subsequently altered. The contact appears concordant although evidence for this is somewhat speculative.

Sample P503/67 (Bottom): TS19991

Location:

430 feet 2 inches - 430 feet 4 inches.

Rock Name:

Coarse-grained, volcaniclastic sandstone.

Hand Specimen:

A medium-grained, dark brownish-grey, massive clastic rock. Minor fractures randomly traverse the rock and are infilled with a secondary carbonate cement.

Thin Section:

A visual estimate of the constituents gives the following:

	<u>%</u>
Igneous rock fragments	60
Quartz	20
Carbonate	5
Clay minerals	10
Hydrated iron-oxides	5

In the main the rock is microscopically massive although there is a slight suggestion of a structure which is defined by the sub-parallel alignment of elongate fragments of quartz. Apart from some of the quartz the clastic components are rounded to sub-angular in shape and have a moderate sphericity. The rock fragments are derived from fine-grained, siliceous, rhyolitic rocks. Many show evidence of having a porphyritic texture. Phenocrysts of high-quartz are commonly included within the rock fragments. Phenocrysts of feldspar are much less common. The most common fragment type is

rhyolitic in composition and composed of a mottled aggregate of alkali feldspar and less commonly plagioclase. Other fragments tend towards the composition of rhyodacites and yet others, although minor, appear to have formed by the alteration and devitrification of an acid glass. Quartz fragments are common and it is possible that many, if not all, were derived from former phenocrysts.

The matrix is composed of hydrated iron-oxides, opaques, chlorite and clay minerals. Occasional flakes of detrital muscovite are found together with minor secondary calcite. Occasional, small fragments of detrital carbonate represent the only obviously sedimentary fragments.

There is a tendency for the grain-size to coarsen with depth down the hole and for the rock fragments to become more complete.

The rock is traversed by a few minor fractures that are infilled with carbonate.

There is no evidence to suggest that this rock may have been subjected to a metamorphic event.

This rock type is very significant in that it petrologically relates this formation to the Corunna Conglomerate. A few fragments of Moonachie Formation - Volcaniclastic phase are preserved in the detritus. These fragments are composed wholly of clastic rhyolitic rock fragments that are cemented together by a little fine-grained sericite. The fragments have been compared with rocks collected by Smale (1966) and are similar in composition and texture. They also appear identical to P526/67 at the bottom of Roopena DDH No.2 to the east of DDH No.1.

History:

Water laid sedimentary rock derived from a rhyolitic area of provenance.

Sample P504/67: TS19992

Location:

431 feet 6 inches - 431 feet 9 inches.

Rock Name:

Coarsely laminated arenaceous siltstone.

Hand Specimen:

A fine-grained, pale-grey, silty rock. The sample has a poor structure defined by variations in the grain-size of the detrital components.

Thin Section:

A visual estimate of the constituents gives the following:

	<u>3</u>
Quartz	35
Carbonate	8
Chlorite	5
Igneous rock fragments	15
Clay minerals	37

The rock is composed of scattered fragments of quartz, carbonate, chlorite and fine-grained igneous rocks the latter of which are identical to those described earlier. A few laths of fine-grained, detrital micas are scattered throughout. Variations in the ratio of fragments to matrix impart a structure which is further enhanced by the sub-parallel alignment of the long axes of the detrital grains. The matrix is essentially composed of clay minerals with lesser amounts of chlorite and very fine-grained quartz grains. In those zones in which there is a deficiency of the coarser-grained clastic components, the orientation of fibrous micas and clay minerals defines the structure. These laths are obviously detrital. All of the detrital components have been derived from a rhyolitic terrain: There is no evidence for any sedimentary fragments.

Sample P505/67: TS19993: TS19994

Location:

432 feet - f32 feet 9 inches.

Rock Name:

Lithic conglomerate.

Hand Specimen:

A conglomerate composed of angular to subrounded rock and mineral fragments set in a fine-grained matrix. Interbedded layers of fine-grained, laminated siltstone have a sub-parallel orientation that defines a prominent structure in the rock. This structure is approximately perpendicular to the core axis.

There is no positive evidence to suggest the rock was composed of igneous rocks or rock fragments.

History:

Sedimentary, deformed and recrystallized.

Sample P508/67: TS19997 - TS19998

Location:

490 feet - 490 feet 2 1/2 inches.

Rock Name:

Fine-grained, acid igneous rock.

Hand Specimen:

A dark reddish-brown mottled rock. The sample is completely massive although there is evidence for incipient deformation.

Thin Section:

The rock is composed of a fine-grained, felsitic mass of feldspar which has been partially reddened. The degree of reddening is variable and imparts the mottled colouration to the rock. The groundmass has a variable grain-size, but there is no consistent variation. Occasional phenocrysts of feldspar are found randomly distributed. Disseminated opaques are common accessory components together with lesser amounts of zircon and apatite. Fracture zones traverse the rock. These have been infilled with a mosaic of crystalline, magmatic quartz and small amounts of opaque minerals. The rock appears to have been subjected to mild deformation immediately after solidification and then been silicified by late-stage magmatic silica. Alteration is too intense to allow adequate identification of the feldspars but it is probable that the sample is rich in potash.

History:

Igneous extrusive.

Sample P509/67: TS19999

Location:

Beda Bore. Torrens 1 mile sheet.

Rock Name:

Pyroxene Basalt.

Hand Specimen:

Dark fine-grained fragments of a massive igneous rock.

Thin Section:

A visual estimate of the constituents from a fresh fragment gives the following:

	<u>%</u>
Plagioclase	60
Opagues	5
Pyroxene	25
Sphene	2
Mica	8

The freshest fragments are composed of fine- to medium-grained subradiating laths of plagioclase with the interstitial areas being occupied by a colourless clinopyroxene (augite). The feldspar is labradorite (An_{50} to An_{60}), crystals of which are partially replaced by muscovite and other colourless micas. Any interstitial areas are occupied by opaque minerals.

The submitted fragments vary in composition, as does the degree of alteration. Some fragments are almost devoid of mafic minerals, and composed wholly of plagioclase feldspar that is partially replaced by carbonate, chlorite and opagues (specular hematite). Although the fresher fragments predominate all gradations between the two extremes are found.

History:

Igneous.

crystals of opaques and angular quartz grains. Clay minerals are very fine-grained and show evidence for diagenetic recrystallization. Occasional mica laths are orientated in the plane of the bedding. Minor amounts of zircon were the only accessories identified. Two opaque mineral phases are present.

The first is pyrite which usually occurs as spherical or sub-spherical crystals which are usually associated with the darker layers in the rock. This implies a syngenetic origin for the pyrite (possibly biological). The second opaque mineral phase is chalcopyrite but this mineral is not closely related to the bedding. In one instance it is clearly related to cross fracturing. Occasionally it replaces pyrite spheroids. The chalcopyrite is definitely not syngenetic but is most likely of hypogene origin. Textural features suggest that the chalcopyrite is not the result of secondary enrichment.

History:

Sedimentary. There is no evidence to suggest a metamorphic event. The recrystallization can be ascribed only to diagenetic processes.

Sample P507/67: TS19996

Location:

477 feet 5 inches.

Rock Name:

A coarse-grained epiclastic lithic greywacke.

Hand Specimen:

A reddish brown, brecciated rock. The sample is completely massive.

Thin Section:

This rock is composed almost wholly of deformed and partially recrystallized quartz-rich sedimentary rock fragments. Each fragment has a pronounced structure defined by planes of deformation along which recrystallization has occurred. Protophase relics of the primary rock type are set in a fine-grained groundmass of recrystallized quartz and some sericite. The primary mineral components are severely strained. It is virtually impossible to distinguish the various fragments. The whole rock after consolidation has been subjected to slight deformation.

Thin Section:

A visual estimate of the constituents gives the following:

	<u>5</u>
Framework - ?Moonable Formation	15
- Acid igneous rocks	50
Matrix	35

This rock compositionally and texturally resembles P503 (volcaniclastic phase). The major differences include a marked increase in the number of fragments interpreted as having been derived from the Moonable Formation - volcaniclastic phase, the coarser grain-size, and the presence of graded bedding.

This rock is the first in which sedimentary rock fragments appear in significant proportions. Fragments of carbonate and deformed and recrystallized quartzites appear.

The siltstone interbeds have sharp contacts with the conglomerate. The fragments of which they are composed are consistent with their having been derived by the weathering of acid igneous rock fragments. Quartz fragments, some of which retain crystal faces, are scattered throughout. Zircon crystals are present in accessory amounts.

History:

Sedimentary.

Sample P506/67: TS19995: PS11251

Location:

465 feet 6 inches.

Rock Name:

Mineralized carbonaceous siltstone.

Hand Specimen:

A dark-grey, fine-grained, laminated siltstone. Sulphide minerals are distributed along parting fractures and bedding planes in the rock.

Thin Section:

The rock is composed of a laminated mass of clay minerals. The laminations are accentuated by sub-parallel precipitations of opaque minerals and/or carbonaceous material. Distributed through the clay minerals are finely disseminated and larger

LATITUDE LONGITUDE

BORE HOLE RECORD

INDEX NO.

39° 42' 51"

157° 24' 24"

NAME OF BORE/WELL: DOOPENA STRATIGRAPHIC D.D.H #1.

GEOLOGIC AREA : BARBER PARTFORM

_____ :

_____ :

_____ :

NOTES ON LOCATION : S. H. dyke to 4th.DATE COMPLETED :

0-430' Doopena volcanic (basalt);

BACKY POINT FORMATION

ABBREVIATED LOG :REFERENCES : Trans. Doq. Soc. S. A. Vol. 94 (1973) p. 209-211 B. Thompson.REMARKS : 70 ft. assays.

463-467 ft vert. depth.
 467-469
 475-483-4
 483-4-490

width.
 4.1
 2.
 5.4
 6.6

% C.
 0.35
 0.53
 0.3
 0.3

(S. H. S. by B. Thompson 8/17/71.)

GEOCHEMICAL RESULTS ROOPENA DDH 1 409

Drilled by DM and analyzed by ASARCO.

INTERVAL (ft)	(PPM)				
	Cu	Pb	Zn	Ag	
0-10	55	190	270	< 1	
10-20	50	340	300		
20-30	40	270	350		
30-40	40	110	190		
40-50	40	135	100		
50-60	30	15	150		
60-70	40	10	90		
70-80	40	50	180		
80-90	45	20	170		
90-100	35	10	140		
100-110	25	15	130		
110-120	30	10	120		
120-130	170	480	190		
130-140	45	430	190		
140-150	45	45	280		
150-160	20	10	180		
160-170	35	15	120		
170-180	30	5	130		
180-190	35	5	110		
190-200	20	10	110		
200-210	35	55	150	↓	
210-220	30	30	160	3	
220-230	45	20	150	< 1	
230-240	40	120	150		
240-250	30	45	430		
250-260	45	15	130		
260-270	40	15	110		
270-280	50	30	120	↓	
280-290	30	35	150	3	

ROOPEENA DDH1
(ppm)

410

INTERVAL (ft)	Cu	Pb	Zn	Ag	
290 - 300	30	160	330	21	
300 - 310	45	150	210		
310 - 320	45	160	190		
320 - 330	35	95	220		
330 - 340	55	340	380		
340 - 350	5	40	600		
350 - 360	35	1100	480		
360 - 370	60	5500	440		
370 - 380	60	1000	500		
380 - 390	5	25	460		
390 - 400	15	25	380		
400 - 410	60	250	280		
410 - 420	100	100	160		
420 - 430	120	15	210		
430 - 440	75	15	120		
440 - 450	30	15	75		
450 - 460	160	50	160		
460 - 470	4300	30	40		
470 - 480	900	160	25		
480 - 490	900	40	25		
490 - 500	85	5	15	5	
500 - 511	40	30	20	<1	



amdel

The Australian Mineral Development Laboratories

Flemington Street, Frewville, South Australia 5063
Phone 79 1662, telex AA82520

Please address all correspondence to the Director
In reply quote: **AN3/316/0 - 3961/72**

411

7 March 1972

The Manager
Asarco (Australia) Pty Limited
323 Wakefield Street
ADELAIDE SA 5000

REPORT AN3961/72

YOUR REFERENCE: Order A1743
IDENTIFICATION: A14701 to A14751
DATE RECEIVED: 23/2/72

Enquiries quoting AN3961/72 to Officer in Charge please.

Analysis by: A.E. Francis

Officer in Charge, Analytical Section: A.B. Timms

ATD
for F.R. Hartley
Director

Roopena No1 DDH.

plm

MAP		STATE	SA
GRID			
PROSPECT	Base Hole Survey		
SAMPLER	SBW	FILE	
TYPE	Core dust		

412

ROOPENA NO. 1.

Diamond drill core
BATCH NO. sub scratch
samples (du)

FORM 6

JOB

3961/12

AMDEL ANALYTICAL SERVICE

TT	Sample No.						Feet		
			Cu	Pb	Zn	Hg	From	To	
1	H 14701		55	190	270	<1	0	10	weather volcan
2	2		50	340	300	<1	10	20	"
3	3		40	270	350	<1	20	30	"
4	4		40	110	190	<1	30	40	"
5	5		40	35	100	<1	40	50	"
6	Str 51						5		"
7	6		30	15	150	<1	50	60	"
8	7		40	10	90	<1	60	70	"
9	8		40	50	180	<1	70	80	volcan
10	9		45	20	170	<1	80	90	"
11	10		35	10	140	<1	90	100	"
12	11		25	15	130	1	100	110	"
13	12x		30	10	120	<1	110	120	"
14	13		170	480	190	1	120	130	"
15	14		45	430	190	<1	130	140	"
16	15		45	45	200	<1	140	150	"
17	16		20	10	180	<1	150	160	? pyrit
18	17		35	15	120	<1	160	170	volcan
19	11 14718		30	5	130	<1	170	180	"
20	12 ✓								

FORM 76

JOB

2961/12

AMDEL ANALYTICAL SERVICE

BATCH NO.

413

2

TT	Sample No.		Cu	Pb	Zn	Fe	From	Act To	
1	FA 12719		35	5	110	<1	130	190	volca
2	20		20	10	110	<1	190	200	"
3	1		35	55	150	<1	200	210	"
4	2		30	30	160	3	210	220	"
5	3		45	20	150	1	220	230	"
6	4		40	120	150	1	230	240	"
7	5x		30	45	430	<1	240	250	"
8	6		45	15	130	<1	250	260	"
9	7		40	15	110	<1	260	270	"
10	8		50	30	120	<1	270	280	"
11	9		30	35	150	3	280	290	"
12	10		30	160	330	1	290	300	"
13	1		45	150	210	<1	300	310	"
14	2		45	160	190	<1	310	320	"
15	3		35	95	220	<1	320	330	"
16	510 51								
17	4		55	340	380	<1	330	340	"
18	5		5	40	600	<1	340	350	"
19	12736		35	1100	480	<1	350	360	"
20	25x								

TT	Sample No.		Cu	Pb	Zn	Fe	From	To	
1	A 14731		60	5500	440	<1	360	370	volcanics
2	8		60	1000	500	<1	370	380	volcanics
3	9		5	25	460	<1	380	390	"
4	51051								
5	40		15	25	380	<1	390	400	"
6	1		60	250	280	<1	400	410	"
7	2		100	100	160	<1	410	420	"
8	3		120	15	210	<1	420	430	
9	4		75	15	120	<1	430	440	arkose
10	5		30	15	75	<1	440	450	"
11	6		160	50	160	<1	450	460	"
12	7	*	4300	30	40	1	460	470	siltstone
13	8		900	160	25	1	470	480	"
14	9		900	40	25	1	480	490	pyrite
15	10		85	5	15	5	490	500	siltstone
16	11 11151		40	30	20	1	500	511 ft.	"
17	45		NO OTHER SAMPLES				Cu Pb Zn - Code C		
18	BLANK.		TAKEN				Ag - Code CV		
19			All Results in PPM						
20									

High Pb - several - Highest 5500 ppm 360 to 370 feet.

High Cu 4300 ppm From 460 to 470 feet.

* DPM records give:

463 to 467	0.35% Cu
467 to 469	0.52% Cu
478 to 483.4	0.3% Cu
483.4 to 490	0.2% Cu

BORE AND WELL SPOIL SAMPLES

Obtained

By

D. KER, 1963.

WESTERN STUART SHELF AREA.

Sample No.	Grid Reference	Latitude: Longitude:	Bore Ref. No.	Name of Bore or Well	Palaeo Sample No.	Remarks
✓ 26502	J/3	28° 29' 138° 41'	23	Apatoongannie Well	F43/63	
26501				1/4 Mile SSW of Mungeranie H/S (Shale)	F44/63	Surface Spoil Sam- ple (not plotted)

WESTERN STUART SHELF AREA.

Sample No.	Grid Reference	Latitude: Longitude:	Bore Ref. No.	Name of Bore or Well	Palaeo Sample No.	Remarks
✓ 26508	E/3	28° 50' 133° 53'	2	Paragon Bore	F12/63	18S, 14N Mt. Willoughby (near Homestead)
✓ 26509	E/3	28° 57' 133° 59'	6	Watchercallem Bore	F11/63	
✓ 26510	E/4	29° 01' 133° 56'	2	Corkscrew Bore	F13/63	
✓ 26513	F/2	27° 57' 134° 27'	9	Timothy Bore	F16/63	Copper Hill Stn.
✓ 26512	F/2	28° 00' 134° 23'	8	Patricia Bore	F15/63	Appreetina South Copper Hill Stn.
✓ 26511	F/2	27° 57' 134° 16'	5	No. 2 Bore	F14/63	Copper Hill Stn. Appreetina South
✓ 26514	F/3	28° 47' 134° 19'	10	No. 5 Bore	F17/63	
✓ 26515	F/3	28° 51' 134° 22'	14	Mt Clarence H/S Deep Bore	F18/63	
✓ 26516	F/3	28° 53' 134° 37'	18	Sputnick Bore	F19/63	
✓ 26517	F/3	28° 29' 134° 11'	31	Pooramungie Bore	F20/63	
✓ 26518	F/3	28° 50' 134° 41'	40	Giddy Giddina Bore	F21/63	

WESTERN STUART SHELF AREA.

Sample No.	Grid Reference	Latitude: Longitude:	Bore Ref. No.	Name of Bore or Well	Palaeo Sample No.	Remarks
/ 26519	F/3	28° 22' 134° 11'	32	Lesley's Bore	F22/63	Murloocoppie West (Willoughby)
/ 26520	F/3	28° 56' 134° 10'	44	Craigs Bore	F23/63	
/ 26521	F/3	28° 55' 134° 15'	45	West Point Bore	F24/63A	
/ 26522	F/3	28° 55' 134° 15'	45	West Point	F24/63B	
/ 26523	F/3	28° 42' 134° 24'	58	Broken Bit Bore	F26/63	
/ 26524	F/3	28° 06' 134° 06'	47	Dead Finish Bore	F27/63	Mount Willoughby
/ 26527	F/3	28° 12' 134° 23'	52	Bridget Bore	F28/63	Evelyn Downs
/ 26506	F/3	28° 50' 134° 27'	16	Yellow Bullock Bore	F29/63	
/ 26505	F/3	28° 57' 134° 39'	29	Willow No. 3 Bore	F30/63	
/ 26504	F/3	28° 56' 134° 20'	41	Abandoned Bore H/S	F31/63	
/ 26525	F/4	29° 05' 134° 36'	129	Box Hole Bore	F53/63	14N, 14S

WESTERN STUART SHELF AREA.

Sample No.	Grid Reference	Latitude: Longitude:	Bore Ref. No.	Name of Bore or Well	Palaeo Sample No.	Remarks
26526	F/4	29° 07' 134° 29'	131	Abandoned	F51/63	
26527	F/4	29° 11' 134° 20'	12	Abandoned Bore	F52/63	Green Sand
26528	F/4	29° 11' 134° 20'	12	Abandoned Bore	F57/63	Blue Sand
26529	F/4	29° 22' 134° 24'	38	Abandoned Bore	F55/63	
26530	F/4	29° 05' 134° 20'	4	No. 2 Bore	F56/63	
26531	F/4	29° 39' 135° 00'	152	Abandoned Well	F50/63	
26532	F/4	29° 10' 134° 16'	11	Abandoned Well	F54/63	Mabel Creek
26503	F/4	29° 03' 134° 16'	3	Abandoned Bedoura Bore	F32/63	Woorong Creek
26533	F/4	29° 06' 134° 28'	130	Abandoned Bore	F49/63	Mount Clarence
26534	F/4	29° 24' 134° 46'	135	Abandoned Bore	F48/63	
26535	F/4	29° 32' 134° 55'	147	Abandoned Bore	F47/63	

WESTERN STUART SHELF AREA.

Sample No.	Grid Reference	Latitude: Longitude:	Bore Ref. No.	Name of Bore or Well	Palaeo Sample No.	Remarks
26536	F/4	29° 02' 134° 18'	180	Ping Pong Bore	F46/63	Woorong Creek
26537	F/5	30° 15' 134° 46'	93	Glynn Bore	F61/63	Bulgunnia
26538	F/5	30° 12' 134° 40'	75	Abandoned Bore	F60/63	Bulgunnia
26539	F/5	° °	94	Sextabyng Well	F59/63	Bulgunnia
26540	F/5	30° 26' 134° 58'	133	New Bore East 34 Pdk.	F58/63	Wilgena
26541	G/4	29° 52' 135° 13'	12	Dept. Works Mirikita	F69/63	13N, 13S The Twins
26542	G/4	° °	15	Barbies Bore	F71/63	The Twins
26543	G/4	29° 58' 135° 08'	18	Coronation Bore	F70/63	
26544	G/4	29° 57' 135° 55'	34	No. 6 Bore	F67/63	
26545	G/4	° °	42	Blue Duck Well	F72/63	Millers Creek
26546	G/4	29° 49' 135° 49'	26	Beetram Downs No. 1	F68/63	

WESTERN STUART SHELF AREA.

Sample No.	Grid Reference	Latitude: Longitude:	Bore REF. No.	Name of Bore or Well	Palaeo Sample No.	Remarks
26547	G/4	29° 05' 135° 42'	60	Pages Well No. 2	F73/63	
26548	G/4	29° 40' 135° 01'	7	New Year Well	F66/63	
26549	G/4	29° 58' 135° 30'	30	Beetram Downs No. 4 Bore	F65/63	
26550	G/4	29° 47' 135° 55'	32	No. 15 Bore	F64/63	Millers Creek
26551	G/4	29° 40' 135° 01'	45	Abandoned Well	F63/63	
26552	G/4	29° 01' 135° 49'	53	New Bore	F62/63	
X 26553	G/5	30° 34' 135° 53'	211	Clemens Bore	F96/63	13S
X 26554	G/5	30° 14' 135° 35'	274	Cavanaugh Bore	F90/63	
X 26555	G/5	30° 17' 135° 47'	278	Birthday Bore	F91/63	
X 26556	G/5	30° 07' 135° 33'	253	Backlands Bore (old)	F100/63	
X 26557	G/5	30° 04' 135° 34'	264	Backlands Bore (new)	F87/63	

WESTERN STUART SHELF AREA.

Sample No.	Grid Reference	Latitude: Longitude:	Bore Ref. No.	Name of Bore or Well	Palaeo Sample No.	Remarks
x/ 26558	G/5	30° 04' 135° 45'	265	Abandoned Gondalloo Bore (old)	F88/63	/
x/ 26559	G/5	30° 18' 135° 48'	279	Buckley Bore	F92/63	/
✓ 26560	G/5	30° 19' 135° 52'	61	Abandoned Bore	F94/63	/
✓ 26561	G/5	30° 07' 135° 49'	57	Paisley Well	F109/63	
✓ 26562	G/5	30° 11' 135° 44'	56	Russell Bore	F101/63	Mt. Eba
✓ 26563	G/5	30° 08' 135° 10'	40	Gondalloo Bore (new)	F93/63	
x/ 26564	G/5	30° 00' 135° 23'	257	Banjo Bore	F89/63	Nth of Mt. Eba
x/ 26565	G/5	30° 01' 135° 23'	258	Twins Stn. Bore	F108/63	
/ 26566	G/5	° °	191	Bore No. 2	F107/63	Not plotted. Near Bore 308 in G/5
x/ 26567	G/5	30° 26' 135° 50'	308	Bore No. 2	F106/63	
x/ 26568	G/5	30° 23' 135° 50'	181	Abandoned Bore Nth. Vivian Well Hut	F96/63	

Sample No.	Grid Reference	Latitude: Longitude:	Bore Ref. No.	Name of Bore or Well	Palaeo Sample No.	Remarks
✓ 26569	G/5	30° 25' 135° 50'	297	Abandoned Bore	F105/63	
✓ 26570	G/5	30° 26' 135° 50'	315	Abandoned "	F103/63	
✓ 26571	G/5	30° 25' 135° 52'	317	Deep Bore	F99/63	
✓ 26572	G/5	30° 25' 135° 50'	314	Abandoned Bore	F104/63	
✓ 26573	G/5	30° 48' 135° 48'	295	Adj. to Pyra- mid Bore	F98/63	
✓ 26574	G/5	30° 57' 135° 39'	237	Abandoned Well (McKinnons Dam)	F85/63	
✓ 26575	G/5	30° 34' 135° 44'	204	Southern Cross Well	F110/63	
✓ 26576	G/5	30° 22' 135° 10'	177	Bluff Paddock Bore	F97/63	
✓ 26577	G/5	30° 49' 135° 27'	154	Sludge	F86/63	
✓ 26578	G/5	30° 24' 135° 50'	193	Police Camp Paddock	F102/63	
✓ 26579	G/5	30° 04' 135° 19'	17	Crombies Bore	F111/63	
✓ 26580	G/5	30° 05' 135° 21'	19	Bald Hill Salt Bore	F112/63	

Sample No.	Grid Reference	Latitude: Longitude:	Bore Ref. No.	Name of Bore or Well	Palaeo Sample No.	Remarks
26581	G/5	30° 11' 135° 15'	24	Jacobs Bore	F113/63	
26582	G/5	30° 14' 135° 14'	26	Sundown "	F114/63	
26583	G/5	30° 18' 135° 15'	27	Mintabyng Bore	F115/63	
26584	G/5	30° 18' 135° 20'	28	Cheadles Bore	F116/63	
26585	G/5		37	Learie Hill Bore	F117/63	
26586	G/5	30° 14' 135° 42'	53	Central Bore	F118/63	
26587	G/5	30° 19' 135° 42'	55	Corkscrew Well	F119/63	
26588	G/5	30° 24' 135° 03'	62	No 1 Pdk Bore	F120/63	
26589	G/5	30° 25' 135° 29'	255	Dump Bore	F121/63	
26590	G/5	30° 02' 135° 25'	259	Twins H/S Bore	F122/63	
26591	G/5	30° 07' 135° 16'	261	Abandoned Bore	F123/63	

Sample No.	Grid Reference	Latitude: Longitude:	Bore Ref. No.	Name of Bore or Well	Palaeo Sample No.	Remarks
26592	G/5	30° 06' 135° 24'	262	Mathesons Bore	F124/63	
26593	G/5	30° 08' 135° 17'	263	New Saltash Bore	F125/63	
26594	G/5	30° 08' 135° 50'	269	Paisley Bore	F126/63	
26595	G/5	30° 10' 135° 32'	272	Boolka Bore	F127/63	
26596	G/5	30° 13' 135° 34'	273	Cavanagh's Bore (1)	F128/63	
26597	G/5		349	Mungillio Bore	F129/63	
26598	G/5	30° 05' 135° 36'	356	Claypan Bore	F130/63	
26599	G/5	30° 05' 135° 10'	357	Sloans Bore	F131/63	
26600	G/5	30° 10' 135° 55'	358	No. 17 Bore	F132/63	
26601	G/5	30° 11' 135° 58'	359	Abandoned Bore	F133/63	
26602	H/4	29° 25' 136° 47'	38	Coward Springs	F124/62	

Sample No.	Grid Reference	Latitude: Longitude:	Bore Ref. No.	Name of Bore or Well	Palaeo Sample No.	Remarks
✓ 26603	H/4	29° 54' 136° 16'	28	5 Mile Well	F80/63	
✓ 26604	H/4	29° 59' 136° 08'	33	No. 1 Bore	F81/63	
✓ 26606	H/4	29° 59' 136° 37'	54	Abandoned Bore (1)	F84/63	Billakalina
✗ 26607	H/5	30° 06' 136° 55'	101	Sth. of Sturt Crk. H/Stn.	F126/62	.13S
✓ 26608	H/5	30° 06' 136° 09'	3	Moodlampie Bore	F134/63	
✗ 26609	H/5	30° 02' 136° 38'	110	Abandoned Bore	F136/63	
✓ 26610	H/5	30° 00' 136° 11'	7	Night March Bore	F135/63	/
✓ 26611	H/5	30° 12' 136° 10'	107	No. 8 Bore	F137/63	✓
✓ 26612	H/5	30° 17' 136° 04'	108	No. 20 Bore Sludge	F138/63	✓
✗ 26613	H/5	30° 00' 136° 37'	109	Abandoned Bore	F139/63	Billakalina
✓ 26614	H/5	30° 08' 136° 12'	111	Woodamulla Bore	F140/63	Billakallina ✓

			NAME OF BORE/WELL:												
Footage		Sample Type	Lithology	Formation	Cu	Pb	Zn	Ag	Mn	U					
Assay No.	From				To	ppm	ppm	ppm	ppm	ppm	ppm				
26501	not plotted	WASHED PALEO SAMPLES	Shale		15	5	25		130						
✓ 26502		"	Unsorted Sand		15	<5	25		170						
✓ 26503		"	Ferruginous Sand.		15	5	95		1000	<2					
✓ 26504		"	Limonic Sand.		5	<5	25		95						
✓ 26505		"	Limonic Shale/sand		25	25	30		270	8					
✓ 26506		"	Poorly sorted sand		<5	5	20		300						
26507		"	Fragments		15	10	35		450						
✓ 26508		"	Poorly sorted Sand.		5	15	30		90						
✓ 26509		"	"		10	<5	25		230						
✓ 26510		"	"		5	5	25		140						
✓ 26511		"	Grey Sand		5	5	85		270						
✓ 26512		"	"		10	10	95		330						

ASSAY TECHNIQUE: AMDEL JOB NO. 2544/72

BORE HOLE ASSAYS WESTERN STUART SHELF AREA. INDEX NO.

			NAME OF BORE/WELL:											
Footage			Sample Type	Lithology	Formation	Cu	Pb	Zn	Ag	Mn	U			
Assay No.	From	To				ppm	ppm	ppm	ppm	ppm	ppm			
26549			Washed paleo sample	calcareous limonitic sand		5	10	40		30				
26550				Poorly sorted sand		5	<5	45		570				
26551				"		90	30	130		90				
26552				Poorly sorted sand + mica		25	15	60		660				
26553				poorly sorted sand		10	5	40		560				
26554				Poorly sorted sand + limonite		5	35	25		95				
26555				"		280	4400	45		240				
26556				Poorly sorted calcareous sand		<5	5	15		80				
26557				Sand + calcareous limonitic		5	5	15		30				
26558				Calcareous sand		<5	5	10		20				
26559				Poorly sorted sand		5	5	15		85				
26560				"		5	5	35		140				

ASSAY TECHNIQUE: AMDEL JOB NO.

2544/72

BORE HOLE ASSAYS WESTERN STUART SHELF AREA

INDEX NO.

			NAME OF BORE/WELL:											
Footage			Sample Type	Lithology	Formation	Cu	Pb	Zn	Ag	Mn	U			
Assay No.	From	To				ppm	ppm	ppm	ppm	ppm	ppm			
✓ 26585			Washed Paleo Sample	Calcareous Sand + limonite		5	10	25		190				
✓ 26586			"	"		5	5	10		20				
✓ 26587			"	Sand		30	10	100		220				
✓ 26588			"	feldspathic Sand + limonite		5	5	15		20				
✓ 26589			"	feldspathic Sand + lignite		25	30	400		2400	3			
✓ 26590			"	"		130	5	160		80				
✓ 26591			"	Calcareous Sand + limonite		15	5	65		30				
✓ 26592			"	Poorly sorted Sand		5	<5	5		5				
✓ 26593			"	Calcareous Sand + limonite		15	5	20		10				
✓ 26594			"	Poorly sorted Sand		<5	5	5		80				
✓ 26595			"	"		5	5	35		40				
✓ 26596			"	Calcareous		5	10	20		25				

ASSAY TECHNIQUE: AMDEL JOB NO.

2544/72

Annual Report on
Special Mining Lease No. 55

Pernatty Lagoon, Woocalla, Mt. Gunson
District of South Australia

July 1965

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ABSTRACT

A geophysical and geochemical survey was performed in the Mt. Gunson district of South Australia. The area of the survey is to the west of Pernatty Lagoon.

The area is 90 miles from Port Augusta and lies 5-10 miles east of the transcontinental railway and the main Adelaide - Woomera road.

The mining district has been previously worked for copper and manganese. The copper and manganese minerals are associated with a sandstone and dolomite contact that follows the western edge of Pernatty Lagoon.

Past production for copper has been 3,250 tons of 8 - 10% Cu (1898-1919) and 32,400 tons of 3.5% Cu. (1941-1943). Past production for manganese has been 34,000 tons of 45 - 55% Mn. (1915-1949).

An induced polarization survey was performed over an area of 45 sq. miles and geochemical samples were taken along the induced polarization lines. At least 21 geophysical and 29 geochemical anomalies were found. At least 12 geochemical anomalies correlate with induced polarization anomalies.

Since definite anomalies in the manganese and copper areas were found, further exploration work is to be proposed.

The presence of widespread mineralization and the positive induced polarization and geochemical anomalies justify a drilling program to identify the anomalies and delineate the potential ore bodies.

A preliminary calculation from the geophysical data estimates the ore reserves in manganese north of Woorcalla as being 820,000 tons. The possibility of a

large low grade copper ore body around Mount Gunson should also be considered. An estimate of the potential in this area could be made from the drilling program. The induced polarization anomalies could be higher grade lenticular copper ore bodies. Ore reserves in this area are 395,000 tons at 3.14% copper.

INTRODUCTION

The following report describes the work and results of an exploration program carried out in the Mt. Gunson district of South Australia from February to April 1964.

The report also includes extracts from earlier publication concerning the geology and past production of the area. These reports are listed in the bibliography.

LOCATION

Special mining lease No. 55 covers an area of 500 sq. miles in the Oakden Hills pastoral run. It is about 300 miles north of Adelaide and about 90 miles north - northwest of Port Augusta. It is bounded to the east by Pernatty Lagoon and to the west by Inland ^{and about} Lagoon. Both the Transcontinental railroad and the Adelaide - Woomera road pass through the center of the lease.

The area is composed of isolated tablelands interspersed with extensive depressions occupied either by salt lakes or alluvium. On the tablelands the vegetation is very scanty with an absence of trees. In contrast the low-lying areas have a well developed covering of mallee trees, except around the margin of the salt lakes.

A few high flat-topped residual hills stand out above the general level of the tablelands. The processs of formation of the present flat land surface, as a result of widespread degradation, is well advanced.

GEOLOGY

Pernatty Lagoon is one of a number of normally dry saline lakes which occupy depressions in the region extending south from the Arcoona Plateau. The region is extensive tableland elevated some 850 ft. above sea level and constituting part of the Stuart Shelf west of Lake Torrens.

A stratigraphic cross section beginning at the surface follows:

- (1) Alluvium occurs as a covering over most of the area.
- (2) Duricrust occurs in the form of a mantle covering the tablelands. The material is composed of a tertiary secondary quartzite and is associated with angular highly silicified fragments of the underlying rocks.
- (3) Upper Sandstones - White Sandstones (100 ft. +) occur as a hard resistant capping on the residual hills. The upper 40 ft. + is a sandstone of variable grain size, not particularly well bedded, but containing hard dense quartzite layer. This formation was probably deposited under shallow water conditions. The middle 30 ft. + is a quartzite sandstone containing small cavities about one half inch in diameter. This pitting is caused by the erosion of uncemented patches of fine sand. The lower 30 ft. + is a cross bedded sandstone suggesting aeolian conditions of deposition.
- (4) Red Sandstones and grits occur under the whole area. Cross bedding is evident and shale layers are interbedded in places with the red sandstones. This rock is generally highly siliceous and in certain locales is copper bearing.

(5) Dolomitic Limestones (50-188 ft.) occur as a persistent stratigraphic horizon on the western shore of Pernatty Lagoon. The Dolomite is of great importance because all of the local manganese and copper mineralization is either distributed in it or in the sandstone beds immediately above or below it. The rock is grey in color and has a blocky, jointed appearance. It represents recrystallized carbonate rocks.

(6) Lower Sandstones (50 ft. +) - These sandstones are similar in color to the red sandstones of the upper series. At Pernatty Lagoon these sandstones are cupriferous immediately below the dolomitic limestone. Shaly layers do occur but they are by no means the predominant rock type.

(7) Rocks at depth - These rocks are of interest because they might represent the source of mineralization in the district. At nearby localities deep bores sunk for water are reported to have intersected igneous rocks. To the south and south-west the nearest porphyritic and granitic rocks outcrop in the region of Iron Knob and the Gawler ranges.

GEOLOGIC STRUCTURE

The Upper Sandstones - The Dolomitic Limestone and the Lower Sandstones form an apparently conformable series of sediments which have a subhorizontal arrangement.

The regional structural setting is one of gentle basins and dome like structures, locally fractured and faulted, but there is no definite evidence of faulting in the district.

In all probability the ore control is the limestone bed. Since copper ore occurs both above and below it, chemical precipitation is suggested as an aid to ore deposition.

MINERAL DEPOSITS

Copper

In the northern part of the district at Mount Gunson, the copper ore is found primarily in the sandstones which are in contact with the dolomitic limestones. The copper minerals associated with the sandstone are found in restricted locales, but in the dolomite traces of copper minerals can be seen over a much broader area. The copper minerals are distributed in the sandstones in an irregular fashion due to the nature of the crossbedding and the changing composition of the sandstone from a gritty texture to a quartzite.

The ore occurs as oxidised, chloride, sulphate, carbonate and siliceous minerals, as well as chalcocite. In a north-easterly direction the minerals become less oxidised. Typical grades in high grade zones are 8-15% and in average zones 3%.

The ore occurrences near Woocalla are very similar to Mt. Gunson. The mineralization is over a smaller area and is not very intense.

Manganese

Manganese is the most common mineral in the Mt. Gunson-Pernatty Lagoon district. It can be detected in the dolomitic limestone wherever this rock outcrops. In an area 5 miles north of Woocalla the manganese ore occurs in sufficient quantities to mine. Elsewhere in the district small patches of manganese ore have been located in workings. Manganese float is noticed over a wide area to the north of Woocalla.

The ore occurs as manganese oxides, chiefly pyrolusite. The grade produced ranges from 35 to 55%. The origin is likely oxidation of the dolomitic limestone caused by surface weathering. This origin is typical of most of the world's manganese deposits.

PAST PRODUCTION

The Mt. Gunson mine is in the northern part of the area. It was first worked from 1898 until 1919. The recorded production for this period was 3,250 tons and comprised dressed ore containing 8-16% Cu. During the period 1941-1943, open cut mining operations of the Zinc Corporation Ltd. produced 32,379 tons of ore averaging 3-5% Cu. Work was discontinued when the minimum grade of 2.5% Cu. could no longer be maintained by open cutting.

The manganese deposits to the north of Woocalla were in production from 1915 until 1949. During this period over 34,000 tons of manganese ore containing 45-55% Mn. was marketed. During 1942 over 8,000 tons was produced.

There are no figures for the production of the copper and manganese mines around Woocalla in the southern part of the district. It is understood to have been less than 500 tons.

TECHNICAL PROCEDURES AND EQUIPMENT

The Induced Polarization equipment used in this survey consisted of the Geoscience Incorporated IP Receiver - Model 5260 and the IP Transmitter - Model 5170 (constant-current, variable-frequency transmitter).

The variable frequency induced polarization method is basically a measurement of the resistance of the ground at two different frequencies. For example .05 and 3 cps. Small differences typically on the order of 5-20%, in the ground resistance are indicative of the presence of electronic conductors. Such conductors include metallic sulphides and graphite but exclude zones of low conductivity such as faults and salt water, where conductivity is due to ionic conductors.

FIELD PROCEDURE

After preliminary trials with different electrode configuration and receiving techniques, the dipole-dipole method was chosen as being the most suitable for the area with a frequency range of 0.3 to 3 cps. The dipole lengths used were 300 ft. and the lengths of the lines ranged from 2 1/2 to 6 miles.

In the northern section of the area where old copper workings are abundant, the traverses were run at approximately 1/2 mile intervals. In the southern section the mineralization is not as well defined. Under these conditions the most efficient use of the IP is to conduct a general reconnaissance survey. To accomplish this the traverses were run at approximately 1 1/4 mile intervals.

It was found that the speed of the survey was considerably improved by skipping a few hundred feet of traverse at every transmitter location. This technique is ideally suited in general reconnaissance prospecting for large, flat lying, low grade deposits which are observed in Pernatty Lagoon. The most likely mineral occurrence in the area is of this type.

Induced polarization measurements were made over a total of 53 1/2 miles of line. Between the Gunyah Lake copper workings and the Big Cut manganese workings, 44 miles of line were run. The remaining 9 1/2 miles are located in the vicinity of the Woocalla workings.

Geochemical soil samples were taken at 1200' intervals mainly along the IP traverses. They were analyzed for copper and the values obtained represent the unit quantities of dithizone used. A total of 290 samples were obtained over 62 1/2 miles of line.

The results of this work are presented in a series of 1" - 200' IP data plots, showing metal factor, frequency effect and resistivity, in a vertical cross section for each line. These sheets are issued separately. The interpretation for each line is listed in Discussion of Results. A tabular presentation of geochemical results is included. A 1": 1 mile base map of the area, displaying the locations of the IP and geochemical lines as well as the positions of the anomalies is also included. Further notes on the explanation of symbols used in these illustrations will be found at the end of this report.

DISCUSSION OF RESULTS

In view of the wide line spacing and general reconnaissance nature of the project, anomalies should be rated and discussed on their own individual merits, even though continuity of anomalous zones between lines can be said to exist in some places.

To facilitate discussion and to provide a ready reference to anomalies, a table of anomalies has been presented. Depth estimates depend a great deal on the resistivity of the medium surrounding the anomalous source; but generally speaking, "shallow" can be taken to mean from the surface down to 300', "medium depth", from 300' to 500', and "deep", below 500'. Ratings are given to anomalies as follows:

- A. Indicates an anomaly having a priority of interest, and may warrant further investigation as discussed in "Conclusions and Recommendations".
- B. May also warrant further investigation but considered to be of secondary importance to A.
- C. Only if further investigation of groups A and B produces encouraging results would further work in this group be feasible.
- D. A group of weak anomalies which have the lowest potential for mineralization. They can be disregarded unless further investigation of the above groups indicate the established presence of a commercially operable ore zone.

PERNATTY LAGOON

LINE	CENTER	DEPTH	F.E.	GEOCHEM.	REMARKS	RATING
33N	73E	deep	1.6	1	May result from more than one source. Eastern half may be partially invalid on account of coupling effects. Strong evidence of a relation to Gunyah workings (sulphides). Weak Geochem. (consistent with depth).	B
21N	48E	-----	---	21 9 2	High geochemical anomaly. Probably near-surface source. No I. P. information available, but moderate anomalies in adjacent lines.	A
9N	57E	shallow	2.3	45	Probably very shallow. Broad. Interesting geochemical association. Possibly related to line 21N geochemical anomaly's source.	A
	138E	deep	1.8	0.5	Deep source consistent with geochemical value. Suspect minor degree of coupling.	B
3S	24E	-----	---	2.5	Moderate geochem. No I. P. information. Isolated.	D
	108E	-----	---	3.5 2 2	Moderate geochem. anomaly. Widespread. Situated West of Gun workings. Interesting location. No. I. P. information.	C
15S	99E	deep?	1.0	0	From pole-dipole trials. Small anomaly-low interest-but may be related to Gun workings.	D
30S	75E	shallow	5.0	3	Most significant I. P. anomaly in Northern group. Near surface effect. Related to copper mined in old open cut (main one) at Mt. Gunson workings. Source probably lies in weak primary mineralization beneath carbonate copper and somewhat to the west of open cut.	A

LINE	CENTER	DEPTH	F.E.	GEOCHEM.	REMARKS	RATING
30S	105E	shallow --med. depth	2.0	0-5?	Geochemical value of 5 @ 600' west of center. I. P. anomaly associated with small open cut near Mt. Gunson homestead.	C
	135E	deep	1.0	0	Ill-defined--no geochemical associations.	D
60S	13E	med.	1.7	0	Well-defined, but weak I. P. No geochem., clean anomaly.	B
90S	135E	shallow	2.5	0	Weak I. P. Not very significant. Values perhaps could be reduced in view of slightly excessive meter drift. Manganiferous outcrop on surface.	D
120S	120E	med.	1.3	0	Weak I. P. but definite anomaly.	B
156S	171E	med.	1.9	0	Anomaly probably extends westward. Weak.	D
216S	180E	deep	1.6	1	Very weak. Slight geochemical association.	D
279S	170E	med.	4.0	no info.	Anomaly reaching moderate size, and extending westward.	C
	180E	med. - deep	3.2	no info.	Anomaly of moderated size, appears to be fairly extensive. Closely related to eastern neighbors. Manganiferous outcrops plentiful--but anomaly does not appear to be due solely to near-surface source.	B
339S	69E	deep	2.5	0.5	Weak, but interesting. Well defined.	C
	100E	shallow --med.	2.9	0-1	Approaching moderate size I. P. anomaly. May extend to east.	B
	117E	shallow	3.4	2	Moderate size I. P. May extend East and West to join adjacent anomaly. Interesting geochemical association.	B

LINE	CENTER	DEPTH	F.E.	GEOCHEM.	REMARKS	RATING
399S	12E	deep	1.9	0	Apparently isolated. Well defined. Probably worth low priority investigation.	C
	96E	shallow --med.	3.5	0	Insufficient information is available to interpret adequately. Both anomalies probably join and may result in more than one polarizable source. Scattered mangiferous float in evidence 87E to 108E.	B
	120E	deep	5.1	0		
459S	66E	deep?	1.6	0	Weak--no geochemical value---probably extends westward.	D
	96E	deep	2.8	0	Approaching moderate size, no geochem., clean anomaly.	B
	123E	med.	3.1	0	Ill-defined, but approaching moderate size. Extends out into lake. Probably connected to above anomaly.	B
519S	18E	deep	1.7	0	Very weak, but clean anomaly.	C
	99E	med.	1.4	0	Weak, but fairly well defined.	C
	132E	shallow?	1.4	0	Ill-defined--low order of magnitude--incomplete information. Extends eastward.	D
	177E	shallow	2.9	0	Very broad anomaly. Results from wide, near surface conductor. Coupling almost certainly present in higher values associated with low resistivities. Anomaly is associated with Big Cut manganese workings. Continuation of anomaly indicated east and west.	B

WOOCALLA

LINE	CENTER	DEPTH	F.E.	GEOCHEM.	REMARKS	RATING
30N	93W	med.	1.9	0.5	Low magnitude anomaly. Weak geochemical association.	C
	75W	med.	2.4	1	Approaching moderate proportions - well defined.	B?
	18W	shallow --med.	0.9	1	Weak anomaly. Almost no polarization evident. Correlation with lines to south.	B
	21E	deep	0.8	1	Weak anomaly, and small. Almost no polarization evident.	D
	75E	shallow	2.3	1	Approaching moderate magnitude and extending eastward.	C
0	90W- 72W	med.?	2.5	3	Moderate sized I. P. anomaly--interesting association with geochemical value. Suspect slight degree of coupling. Small Mn. workings in area.	B
	60W	shallow	2.0	0.5	Approaching moderate size--but not very significant.	C
	45W	deep	0.9	3	Weak anomaly, but possesses moderate geochemical association.	C
	12W	shallow --med.	1.4	7	High geochem. value with low-magnitude I. P. anomaly. Anomaly close to sandstone-dolomite contact. Old Sweet Nell workings in vicinity.	A
	6E	?	1.3	?	Extends westward and probably related to its neighbor.	B
	30E	?	0.9	2	Incomplete anomaly with a true center probably around 24E. Moderate geochemical value.	C
	75E	?	1.3	no info.	Incomplete--extending eastward. Approaching moderate size.	C

LINE	CENTER	DEPTH	F.E.	GEOCHEM.	REMARKS	RATING
WS	6E	med. -- deep	3.8	1	Moderate anomaly situated in general area of anomalous values. Probably arises from more than one source.	A
	24E	shallow	3.3	no info.	Moderate anomaly near surface. Close to railroad.	A
	72E	shallow --med.	1.4	11	High geochem. with low order I.P. anomaly. May be related to similar anomaly on line 0. This area between the anomaly and Sweet Nell Mines worth further investigation.	A

POSSIBLE PRODUCTION CONSIDERATIONS

Copper

Estimates of tonnage for the northern area are about 395,000 tons of ore with an average grade of 3.14% Cu. These reserves were calculated for the Zinc Corporation Ltd. and can be regarded as being a fair estimate.

The major mineralization occurs in an area of about .5 sq. miles. The ore reserve of 395,000 tons by no means covers this whole area. The induced polarization and geochemical data increase the potential area to about 2-3 sq. miles. It is impossible to calculate an exact estimate of potential tonnage but it is reasonable to assume that this area could contain upwards of 20,000,000 of ore per square mile taking an average depth of 10 feet for the ore body. The ore body has been defined as the rock zone containing a grade greater than 3%. It is apparent that ore values of 1% + are contained in the surrounding rock and the potential of the area as a low grade copper deposit would be increased if these values were known. The induced polarization anomalies in this area could represent higher grade copper which is as yet unexposed.

It is possible that 200,000,000 tons of .5-1% copper exists interspersed with zones containing higher values. A continuing exploration program would determine the extent of the mineralization and the grade of the potential ore.

Manganese

The manganese deposits 6 miles north of Woocalla are about 10 feet thick and are under about 4 feet of alluvium. They form large ore bodies with ill-defined boundaries with a grade varying from 25 to 55% manganese. An average dressed sample would have the following analysis:

<u>Element</u>	<u>%</u>
Manganese	51.00
Iron	2.80
Phosphorus	.05
Sulphur	.10
Insoluble	2.50

A rough estimate of ore reserves based on the geophysical data follows.

The following values have been used in this calculation:

1 acre-foot = 43,560 cubic feet;

density of 10% Mn. ore = 3.0;

12 cubic feet of rock = 1 short ton;

from which is derived that:

1 acre-foot contains 3630 short tons of rock;

363 tons of 10% ore; containing 50% Mn.

The following assumptions are made to arrive at a conservative estimate of indicated ore:

- (1) Only shallow IP anomalies related to manganese in outcrop, float, or nearby mine are considered (5 in all).
- (2) The anomaly represents a flat-lying ore body with a square outline which has been cut parallel to one side by the traverse line.
- (3) The rock contains 10% manganese oxides averaging 50% manganese metal.

Anomaly (1)	Geoscience Rating (1)	Length Feet(1)	Probable Acres (2)	Possible Acres (3)	Tons of 50% ore per Acre-foot
90S/135E	D	790	3.8	11.9	4280
279S/180E	B	1320	6.35	40.5	14600
399S/96E	B	1630	7.8	61.0	28000
519S/177E	B	950	4.6	21.2	7640
0/90W-72W	B	1580	7.6	58.0	274000
TOTAL -					81920

(1) From the list of geophysical anomalies

(2) A width of 210 feet and length equal to anomaly length.

(3) Taking anomaly length as one side of a square deposit.

The indicated reserve of manganese metal is 82,000 short tons per foot of depth. If a working thickness of 10 feet is assumed, the indicated ore reserve is 820,000 tons having a dollar value of about \$21,000,000.*

* The London Metal price for Indian Manganese, 46-48% was about \$26.00/ton, exclusive of freight charges. (March 5th 1965)

CONCLUSIONS AND RECOMMENDATIONS

The mining district of Mt. Gunson - Pernatty Lagoon is subdivided into three separate anomalies areas. The first is the copper area in the north near Mt. Gunson. The second is the manganese area in the middle near Big Cut. The third is the copper and manganese area in the south near Woocalla.

These areas have a number of geophysical and geochemical anomalies and they are listed below.

(1) The Northern Area (lines 33N-60S) - The correlation of anomalous I. P. responses, geochemical indications, and the old copper workings is encouraging, especially in view of the accepted fact that mineralization in the region is disseminated and therefore more difficult to define. The I. P anomalies are of a magnitude consistent with low grade mineralization, and in general, we feel that this northern area has an interesting potential for a low grade copper prospect.

(2) The Middle Area (lines 339S-519S) - The I. P. results have delineated an extensive belt of weakly polarizable material that seems to lay mainly in the dolomitic limestones, close to the sandstone dolomite contact. The presence of old manganese workings, and the notable lack of geochemical copper trace in the area (with the exception of line 339S) leads us to believe that this anomaly results from a broad low grade band of mangiferous material within the dolomite. Cursory outcrop sampling in the vicinity of anomalies lends support to this view. The possibility of exploitable concentrations of manganese existing within this band should not be overlooked.

(3) Southern Area (Woocalla) - Again, there is an interesting combination of a geochemical response with anomalous I. P. values (line 0) in the vicinity of an abandoned copper prospect (Sweet Nell). Similar anomalies are observed to the south, and, as in the northern area, low grade mineralization is probably the origin of the I. P. response.

A large group of anomalies to the west displays a striking similarity from line to line, and this would indicate a degree of continuity of the source. Manganese has been worked here in the past, and an old copper mine exists some distance along strike to the south in the same geological setting.

This southern area has encouraging possibilities for the discovery of further manganese and copper deposits.

The locations of recommended subareas within each of the major areas just mentioned are presented below.

	<u>Location</u>	<u>Center of Anomaly</u>
Northern Area	33N	73E
	21N	48E
	9N	57E
		138E
	30S	73E
Southern Area	30N	75W
		18W
	0	90W-72W
		12W
		6E

	<u>Location</u>	<u>Center of Anomaly</u>
Southern Area Cont' d	WS	6E
		24E
		72E
Middle Area	339S	100E
		117E
	399S	96E-120E
	459S	96E
		123E
	519S	177E

Earlier geologic investigations of the area suggested further exploration work should be conducted. The presence of widespread mineralization which is visible over large portions of the area as float and outcrop supports this view. The fact that highgrade 10-15% copper and 35-55% manganese was mined in the past in areas where the ore was exposed near to the surface, would suggest the presence of further occurrences in locations which have not been eroded as much if favorable geophysical and geochemical results were obtained.

The geophysical and geochemical survey performed in this area, is the first logical step in a program of geologic exploration. In the Pernatty Lagoon - Mt. Gunson district, the exploration program has shown definite anomalies within three areas.

Considering the potential production of the area, based on geophysical data and geologic interpretation, the exploration program in the district should continue. A drilling program is definitely justified in two areas. They are the copper area around Mt. Gunson and the manganese area

north of Woocalla. It is suggested that these areas be investigated first, since the area around Woocalla is a secondary target in terms of an overall exploration program. It should be investigated though, when the work in the other areas nears completion. In fact if a deposit is proved in the district, it is imperative that further exploration be performed in all adjoining areas.

A drilling program would serve primarily the following purposes:

- (1) It would identify the anomalies.
- (2) It would give the grade of ore present.
- (3) It would delineate the size of the bodies and show whether the anomalies are contiguous from line to line.
- (4) It would provide valuable structural data as to the nature and the cause of the mineralization.
- (5) It would provide ore reserve figures which are essential in any future evaluation of the areas in terms of mines.

Initially, a more detailed geologic study should be made, so as to correlate the geophysical data with the geology. From this study the location of drill sites can be made. It would be most useful to study further, the geochemical samples and obtain ppm analysis for copper. It would then be possible to ascertain the surface expression of copper. This data would further assist the location of drill sites.

The ore calculation in the manganese area north of Woocalla are based on possible zones around the lines and do not represent an ore body running from line to line. If in fact these anomalies did run from line to line a very large manganese deposit would result. It would be improper to predict a potential tonnage if this was the case, but 820,000 tons of manganese as an initial figure, justifies the drilling program in this area.

The ore calculation in the Mt. Gunson copper district are based on earlier reports. An ore reserve of 395,000 tons of 3.14% copper will not make a major mine but the presence of wide spread disseminated oxidised copper and the numerous anomalies within an area of 3 sq. miles, justifies a major exploration effort in this district.

A drilling program was carried out, between Mt. Gunson and Gunyah Lake, in the past. It was designed to prove the continuity of the ore bearing horizon within the area. It supposedly eliminated the possibility of an extensive highgrade deposit. It is apparent though, from an analysis of the geochemical data, that the drilling was performed along a line having a low surface expression of copper. At Gunyah Lake the anomalous copper zone is slightly to the west and has an area of about .5 sq. miles. At Mt. Gunson the major induced polarization response and geochemical surface expression is about 5 miles to the east, within an area of about 1.5 sq. miles. It is also apparent that this program was not concerned with the low grade (1% Cu.) potential of the district.

It is proposed that another drilling program be performed with the object of ascertaining whether a large, disseminated, low grade copper deposit exists.

APPENDIX - Explanation of Symbols Used in Illustrations

Established Anomaly - This symbol represents the surface projection of a zone over which anomalous I. P. values were encountered. It is not intended to indicate the physical limits of the polarizable source. The symbol has been drawn in either on upper or lower position on the data sheet, according to whether the anomalous values occur mainly in the first (shallow source) or second (deeper source) dipole separations.

Doubtful Anomaly - The above remarks also apply, with the additional qualification that the symbol indicates values which

- (a) are due mainly to electromagnetic or capacitive coupling effects,
- or (b) may be false effects from excessive instrument drift,
- or (c) are ill-defined or weak.

An arrowhead at the end of an anomaly symbol indicates that the anomaly may have a possible extension in that direction.

Geochemical analysis number. This is a qualitative guide only to the presence of copper in a surface soil sample, and is not a direct linear measure of the amount of copper present.

The Contours - Resistivity values in Ω -meters are plotted above the line and metal factor values below. In both cases, the values from the closest dipole separation ($n=2$) are plotted nearest the line. Contour interval is logarithmic. The frequency effect is shown in parentheses below the metal factor value. This represents the percent voltage change from 3 cps. to 0.3 cps.

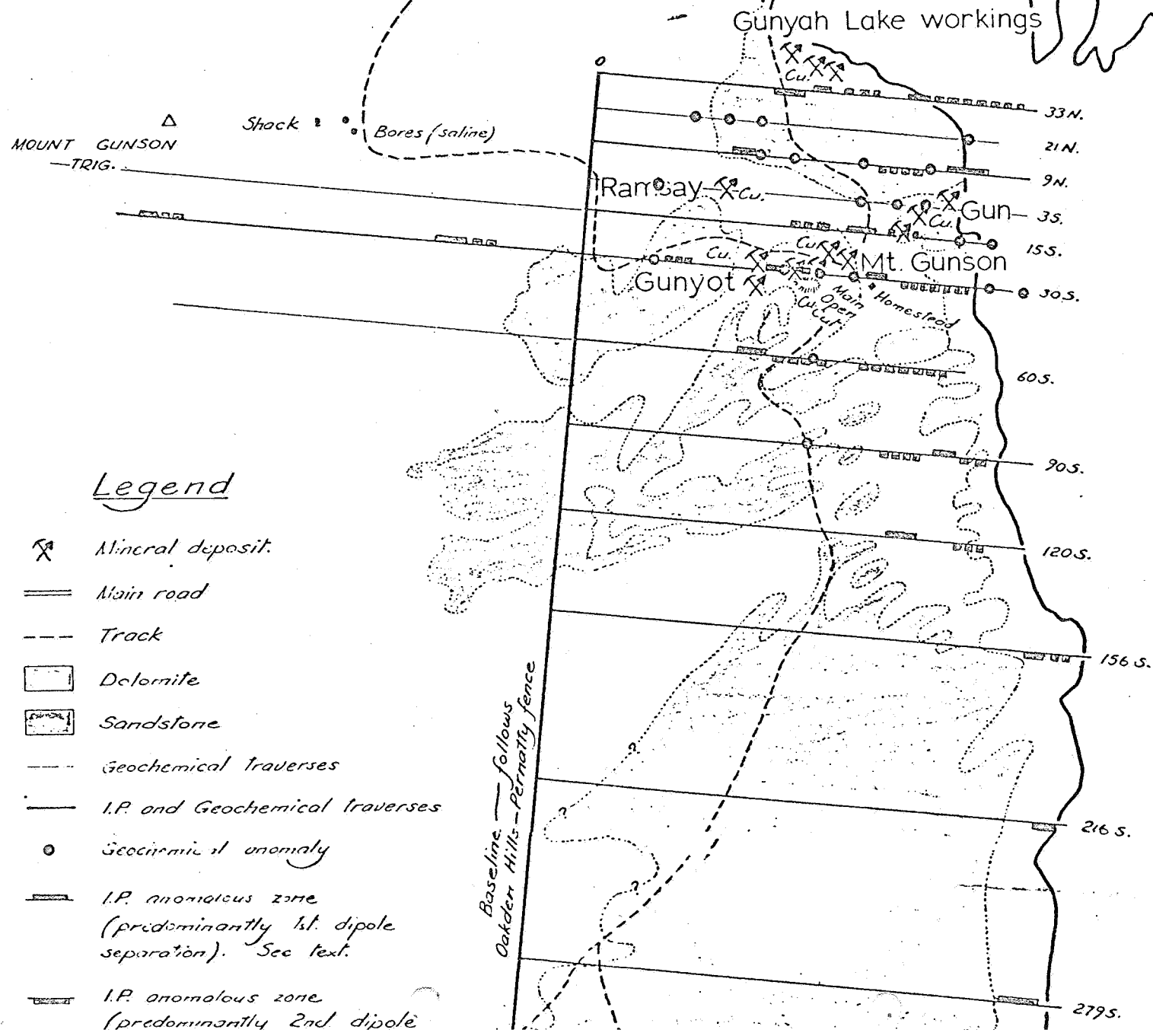
Location of sample having a copper value number greater than 1.

Line	Stations west																Stations east																
	120	168	156	144	132	120	108	96	84	72	60	48	36	24	12	0	12	24	36	48	60	72	84	96	108	120	132	144	156	168	180	192	
Pernatty																																	
33N.																	0	0	0	0	0	0	1	1	1	0	0	0	1				
21N.																	1	0.5	0	2	2	2	0.5	1	1	1	0	3	0				
9N.																	0.5	0	0	0	0	0	3	0	2	0.5		0.5	3				
3S.																	0	0	2.5	0	0	0	0.5	1	3.5	2	2	0.5	1	0.5			
15S.	0	1	0.5	0	0	0	0.5	0	1	0	1	0.5	0.5	0	0	0	0	0	0	0.5	0	0	0.5	0	0	0.5	0	2.2	2.2				
30S.				0.5	0.5	0.5	0	0.5	0	0.5	0	0.5	0	0	0	0	0	0	2	0	0	0	3	3	5	0	0	0	6	3.5			
60S.				0	0.5	0	0.5	0	0	0	0	0	0	0.5	0	0.5	0.5	0	0	0	0	1	1.5	0.5	1	1	0.5	1	1				
90S.																	0	0	2	0	0	0.5	0	0	0	0	0	0	0	0.5	1		
120S.																		0	0	0	0	0	0	0	0.5	0	0	0	0	0	0		
156S.																	0	0	0	0	0	0	0	0	0	0	0	0.5	0	0	0	0	0
216S.																	1	0	0	0	0	0	1	0.5	0	0	0	0.5	0	1	1	1	1
279S.																		0.5	0.5	1													
339S.																		0	0	0	1	0.5	0.5	0	0	1	1	1					
399S.																		0	0	0	0	0	0	0	0	0	0	0					
459S.																		0	0	0	0	0	0	0	0	0	0						
519S.																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
579S.																		0	0	0	0	0		0	0		0						
Woocalla																																	
30N.						0	0.5	0.5	1	1	0	0.5	1	0	1	1	0.5	1	1	1	0.5	1	0.5										
0									3	1	0.5	3	0.5	3	1		1		0.5	0.5													
WS.																	1		0	1	0.5	1											

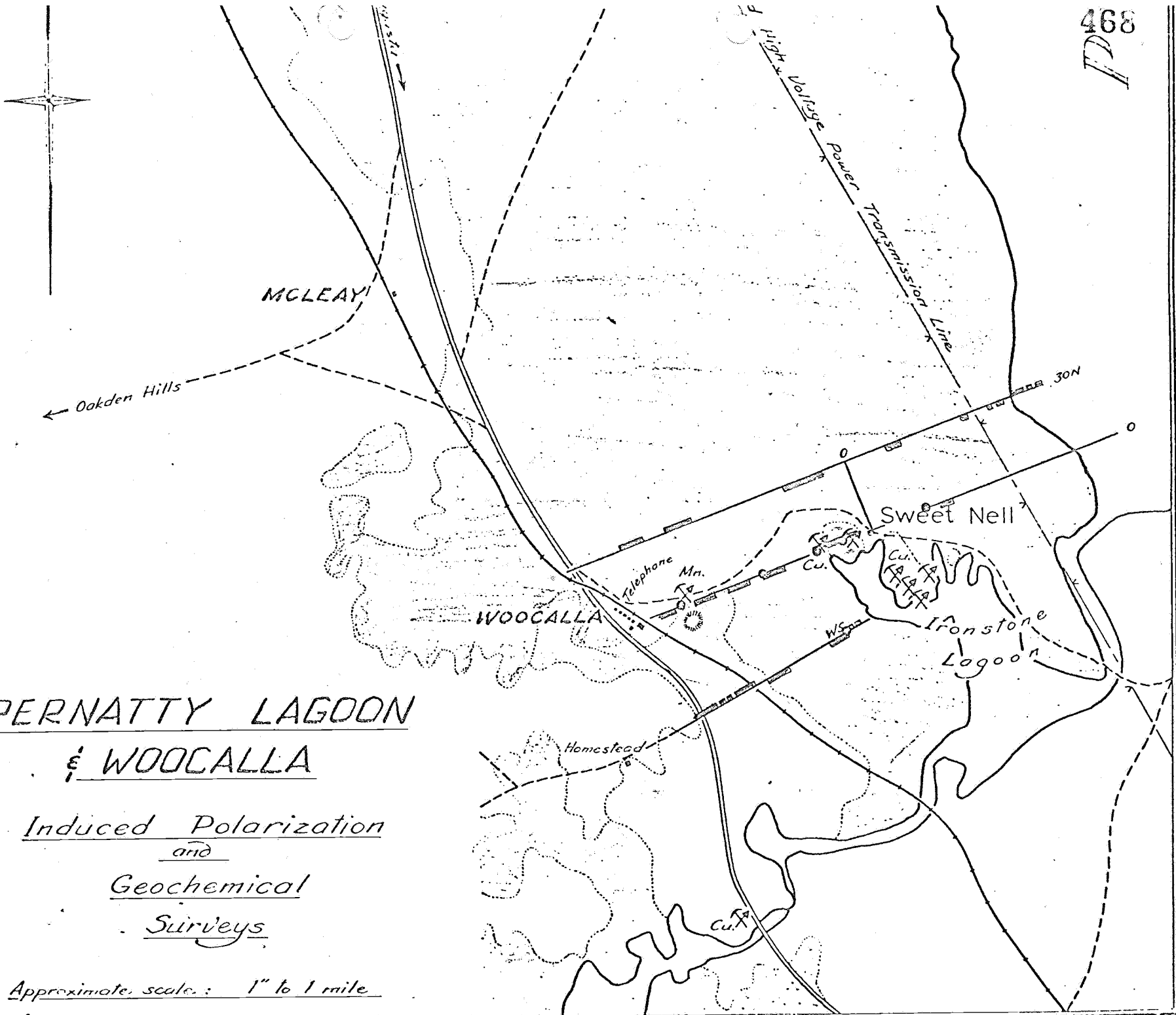
Notes:

Values express unit quantity of dilution used in analyses. Quantities generally increase with increasing incidence of Cu in sample, but this relation is non-linear. The table serves only to point out anomalous conditions in the surveyed area. Significant values above

Pernatty Lagoon & Woocalla
TABLE of GEOCHEMICAL
VALUES



2100.6

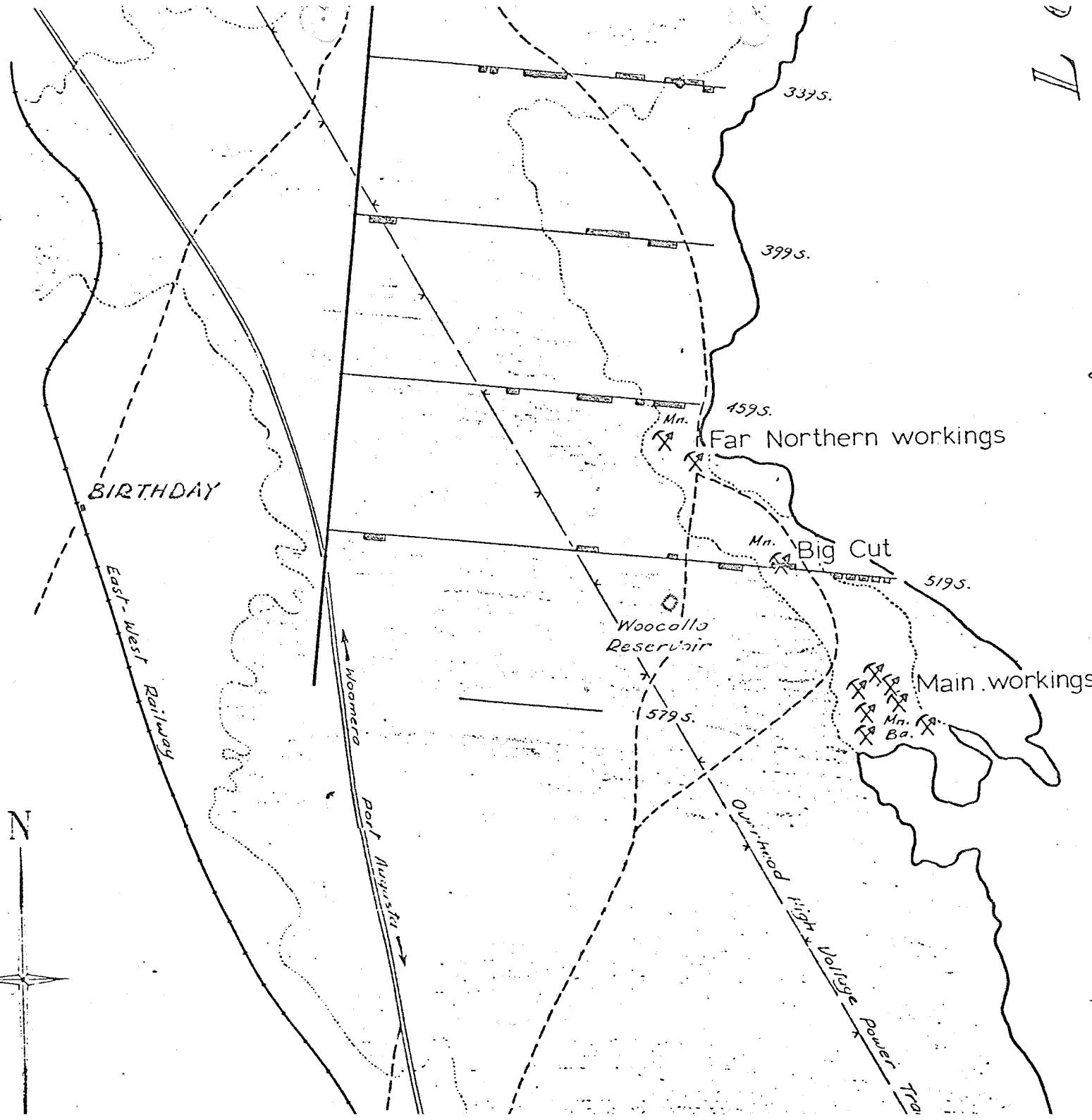
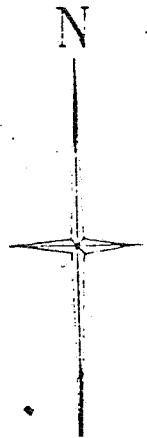


PERNATTY LAGOON
& WOOCALLA

Induced Polarization
and
Geochemical
Surveys

F.I.

Approximate scale: 1" to 1 mile



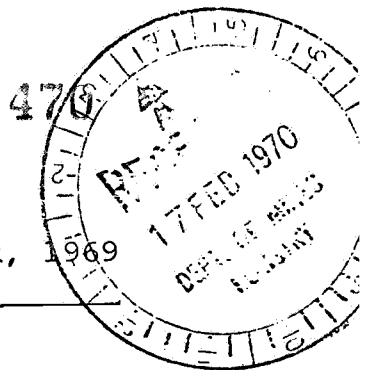
Peel at t y

THIN SALT CRUST
DRY CLAYPAN

315
S.M.L. 135

PERNATTY LAGOON

REPORT FOR THE SIX MONTHS ENDING 15th DECEMBER, 1969



Exploration in S.M.L. 135 is directed to discovery of mineral deposits similar to those exposed in the adjacent areas of -Pernatty Lagoon. The latter deposits, which include copper, manganese, and barytes ores, are contained in sub-horizontal sandstones, shales and dolomites of the Whyalla Sandstone Woocalla Dolomite and Pandurra Formation. The regional geological data currently available on these formations indicate they are disconformable, and the disconformities are loci for mineral concentration. It is only in the narrow zone where the mineral deposits outcrop, on the west side of Pernatty Lagoon and north of Lake Dutton, that erosion has exposed all three formations. Into S.M.L. 135 the Woocalla Dolomite and Pandurra Formation pass into the subsurface and only sporadic exposures of the Whyalla Sandstone are visible in the present land surface.

It is anticipated similar conditions of stratigraphy and structure favouring the concentration of mineral deposits as are exposed at Pernatty Lagoon may be present in the shallow subsurface in areas of S.M.L. 135. Operations for the six months of the least to 15th December 1969 designed to locate such conditions have consisted of geological reconnaissance and literature surveys, a refraction seismic survey, drilling operations and geochemical assay of drill cuttings.

The geological reconnaissance and literature surveys confirmed earlier conclusions that the sand cover which blankets most of the areas of interest limits the application of surface geological and geochemical techniques. Early emphasis therefore has been placed on geophysical and drilling operations with geochemical control through analysis of drill cuttings. All these latter operations have been executed under contract

with Department of Mines equipment and personnel, overall geological supervision being provided by the Company.

A refraction seismic survey was carried out between 4th-23rd August. A Texas Instruments 7000B 24 channel recorder with attendant equipment and personnel was used to complete 35 spreads on an approximate 4 square mile grid pattern over an area of some 700 square miles. A report on the survey by Mr. I.S. Rowan, supervising geophysicist, has been submitted (reference Dept. of Mines Rpt. Bk 69/80, G.S. 4334, D.M. 806/69.)

In summary velocity sets were recorded over the subsurface which corresponded to those obtained over outcrops of the Whyalla Sandstone, Woocalla Dolomite, and Pandurra Formation. Results indicate that in a number of areas onlap-offlap conditions are present in the shallow subsurface between the formations and structural variations are superimposed on the east trending gentle regional dip. As might be expected precise correlation between outcrop and subsurface velocity sets is lacking. It is anticipated that such problems of interpretation will be resolved as subsurface data accumulates during progress of the follow-up drilling operations.

With the seismic survey results as a guideline the drilling operation was planned for drill holes to intersect the Whyalla Sandstone and Woocalla Dolomite and terminate in the top part of the Pandurra Formation, geochemical analyses to be made of the drill cuttings and the results of the geochemical work to be used as a guide to specific target areas for drilling. A caravan mounted atomic absorption spectrometer was provided in field for the analysis of drill cuttings. Problems associated with the drilling equipment have retarded progress on this operation. The initial rig allocated to the programme drilled five holes for a total footage of 357', between 26th - 28th August, the drill did not penetrate to the desired objective and was replaced with a Failing W.W.I. rig with a larger compressor and capacity to hammer drill the hard

Page 3.

sandstones encountered. The Failing W.W.I. commenced drilling on 19th November and had drilled 11 holes for a total footage of 1,246 feet by 27th November. On the latter date a cylinder in the compressor collapsed necessitating major workshop repairs. It was decided therefore to suspend operations during mid-summer and recommence drilling in 1970 when cooler weather conditions prevailed. Details of the drill holes and geochemical analyses on operations completed to date are held by the Geological Survey and Drilling Branch. Interpretations of the results are to be incorporated in the report compiled when the drilling operation is completed.

Bore 7.

- 0- 98 feet Red and yellow clayey sand.
- 98-303 feet Cream sand, mostly fine.
- 303-323 feet Yellow and pink fine sand.
- 323-390 feet Fine cream sand.
- 390-408 feet Pink medium sand.
- 408-435 feet Cream and yellow medium sand.
- 435-520 feet Fine pink sand.
- 520-527 feet Pink sand and ironstone-laterite.
- 527-540 feet Dark grey black pyritic and carbonaceous sand.
- 540-567 feet Fine, dark grey carbonaceous silty sand with fine, angular to subangular quartz grains, pyrite, biotite.
- 567-595 feet Grey-white kaolinized sandstone and clay (?Permian).
- 595-740 feet Green pyritic siltstone interbedded with sandstone and conglomerate.
- 740-1721 ft. Chocolate and brown siltstone and shale.
- Below 959 feet the formation belongs to the Marazion Series.
- 1721-1724'6" Diorite, believed to be basement.

Bore 14.

- 0- 68 feet Red brown sand and clayey sand.
- 68-202 feet Chocolate and blue-grey siltstones and shales and brown sandstones (Upper Proterozoic).

Bore 15.

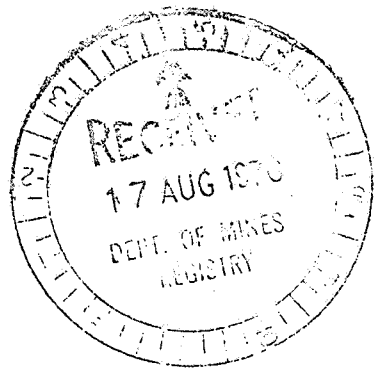
- 0-114 feet Dark red sands and clayey sand.
- 114-126 feet Yellowish-brown sticky gritty clay with quartz grains black carbonaceous and pyritic aggregates, shale fragments. The age is uncertain, but the black aggregates may represent weathering or reworking of the Eocene paralic clays.
- 126-402 feet Greenish grey and chocolate siltstones and sandstones (Upper Proterozoic).

Bore 16.

- 0- 78 feet Red and yellow terrestrial sands.
- 78- 87 feet Yellow-brown clay of uncertain age.
- 87-227 feet Blue and chocolate siltstones and sandstones.

Bore 17.

- 0- 80 feet Red brown clayey sand and kunkar.
- 80-103 feet White sandstone - age undetermined.
- 103-220 feet Red brown and white siltstone, chocolate, brown and green shale and dolomite.



SPECIAL MINING LEASE 315 - STUART'S BLUFF MINERALS LTD.

PERNATTY LAGOON, SOUTH AUSTRALIA

REPORT ON EXPLORATION - 30TH APRIL, 1970.

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INTRODUCTION:

Special Mining Lease 315 was issued to Stuart's Bluff Minerals Ltd. for a term of one year on 16th June, 1969. Lease conditions require the Company to undertake geophysical, geochemical and drilling operations at minimum expenditure levels of \$10,000 in the first six months and \$15,000 in the second six months. The Company has completed its work obligations and exceeded the minimum expenditure requirements for the first six months to December 1969. It is to complete a combined shallow drilling and geochemical operation by 16th June, 1970, to have fulfilled its work obligations for the second six months.

Operations during 1969 consisted of a seismic survey and a combined drilling and geochemical operation, carried out under contract by units of the South Australian Department of Mines and, in conjunction with this work, a geological reconnaissance and literature study by the Company's geologist. The seismic survey, geological reconnaissance and literature study are completed. The combined drilling and geochemical operation, initiated in the last quarter of 1969 was suspended on 27th November following drilling equipment breakdowns, with the intention of resuming the work in 1970. The programme consists of drilling approximately 50 holes to an average depth of 100' with the drill cuttings being geochemically analysed with a field mounted atomic absorption spectrometer.

This report covers the objectives and results of exploration by the Company in the lease area to 30th April, 1970.

LOCATION DATA:

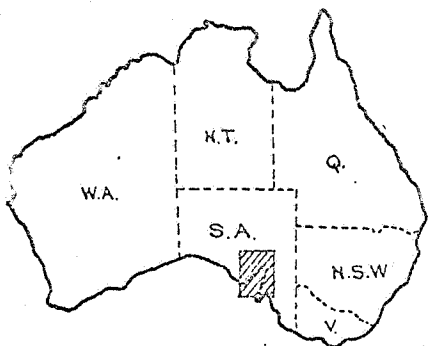
SML 315 covers 840 square miles of South Australia between Port Augusta and Woomera; with Lake Torrens lying east of its eastern boundary and its western boundary passing through the eastern side of Pernatty Lagoon.

The dissected plateau of the Andamooka Range, 600' - 800' above sea level with its surface pitching south to north at a low angle, bounds the east, north and north-west sides and extends into the north-east corner of SML 315. Below the plateau to the south and west lies a sand covered plain, 200' - 300' above sea level in the lease area, falling gradually south-east to sea level around Port Augusta. Salt pans, dry or forming shallow ponds of saline water, and sand dunes are common land forms of the plain. Poorly exposed sandstones form minor undulations to low hills rarely rising to 50' above the level of the plain.

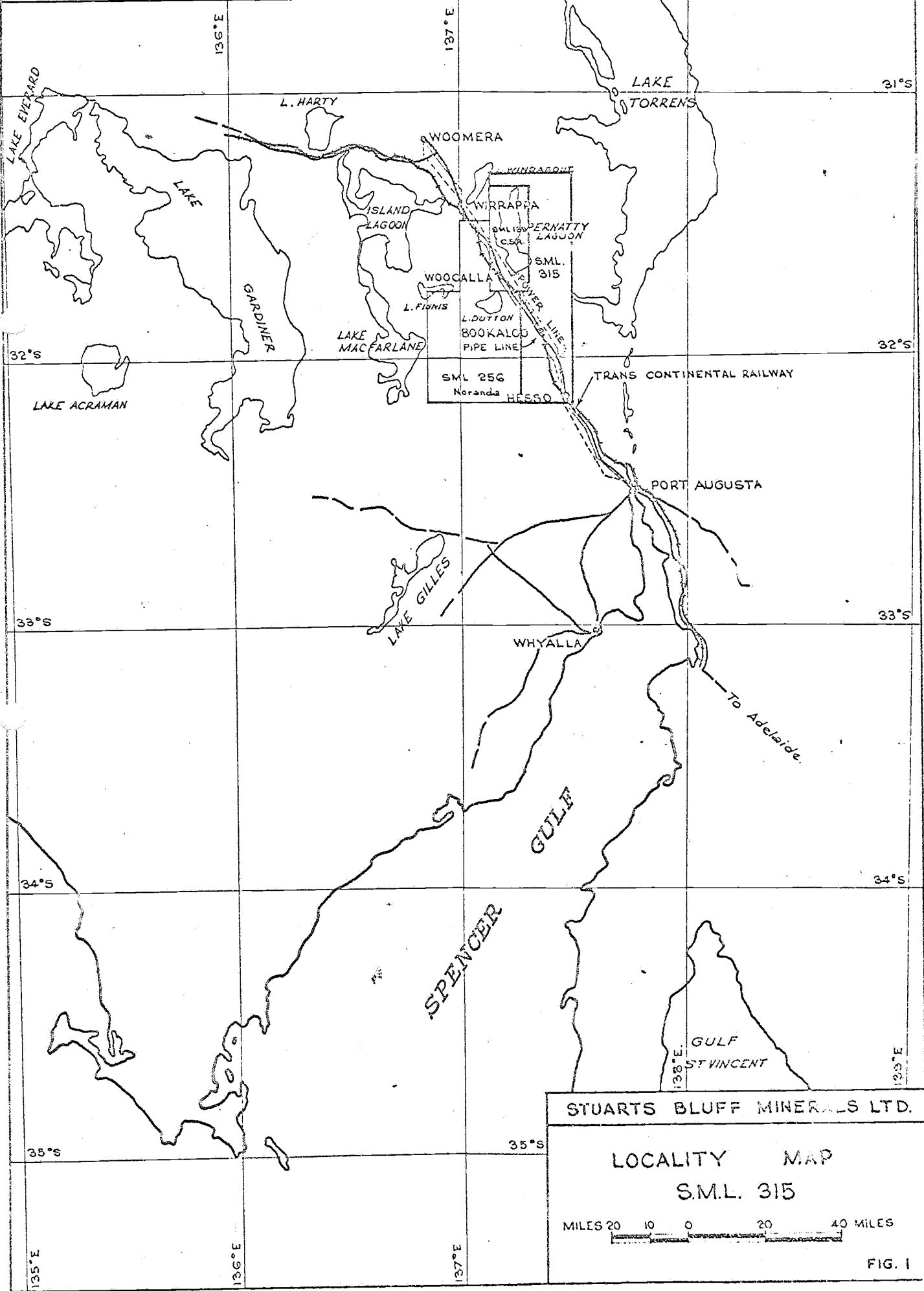
Climate of the region is semi-arid. Rainfall is low and unreliable, the average annual over the last twenty years approximating 7 inches. Evaporation rate is high. Summer temperatures frequently exceed 100°F, average maximum approximately 90°F, with the average winter minimum approximately 45°F. Saltbush and bluebush shrubs, the stunted Mulga and Myall trees and localised developments of She-oak are the dominant vegetation.

Grazing of sheep and cattle constitutes the principal industry of the region. Copper, manganese and barytes ores, located along the west side of Pernatty Lagoon, have been spasmodically mined in the past and small tonnages of ceramic clay have been quarried at Woocalla. Mining of the copper ores is currently being rejuvenated.

The Stuart Highway and the Transcontinental Railway, linking Adelaide and Perth, pass diagonally through the southern half of SML 315, approximate distances by these routes from Port Augusta being 30 miles to the southern and 60 miles to the western boundaries of the lease. An 8 inch water pipeline and a high tension power transmission line, connecting Woomera and Port Augusta, run parallel and adjacent to the road and rail systems. An all weather road runs northward through the lease area from Bookaloo to Pernatty and Yeltacowie Homesteads. Numerous tracks traverse the lease area.



LOCALITY PLAN



Copper with associated silver, manganese and barytes deposits are partly exposed along the west side of Pernatty Lagoon where they occur as tabular, almost horizontal bodies in sandstones, dolomites and shales of Pre-Cambrian age (figure 3).

The copper minerals (including malachite, chalcocite, covellite, bornite and atacamite) occupy interstices in the porous sandstones, cracks in the dolomite and impregnate the black shales associated with the dolomite. Thickness and grade of the ores vary, most are low grade, with worked thickness up to 14 feet in the sandstone sections and 8-15% typifying high grade zones.

The manganese ore results from enrichment of manganiferous dolomite. The ore is uniformly distributed over appreciable areas of dolomite and also occurs as irregular pockets along joint planes. The abandoned workings were about 14 feet deep over an area of approximately an acre. Crude ore produced contained 35-57% manganese.

The barytes is closely associated with the manganese deposits and occurs as massive, coarsely crystalline aggregates. The main exposure, the vertical dimensions of which have not been tested, occupies a rise slightly above lagoon level, approximately 150 feet wide and 1500 feet long. A fairly pure grade ore was derived from the now abandoned workings.

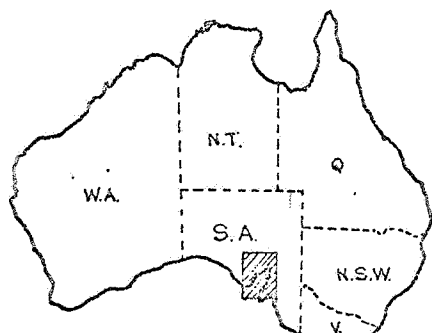
Since discovery of the deposits mining has been spasmodic and production limited. Between 1898-1937 recorded output of copper ore amounted to 3,250 tons, mainly dressed ore containing 8-16% copper. Between 1941-1943 32,380 tons of copper ore, averaging 3.5% copper, were mined for a yield of 1,084 tons copper and 14,523 ozs. silver. In the last two years C.S.R. Co. Ltd. and associates have been preparing sections of the northern copper deposits for open-cut mining. Milling, crushing and flotation equipment is being erected and initial production of concentrates is planned for 1970. Current estimate of ore blocked out for open-cut mining and treatment is 3,200,000 tons containing about 23 lbs. copper and 0.35 ozs. silver per ton. Planned average annual production of concentrates over a ten year period, expressed in terms of copper metal equivalent, is 2500 tons, plus associated silver.

Recorded production of manganese ore, from 1915 to 1949 when mining ceased, amounted to 34,000 tons containing 40-55% manganese. Production of barytes, over the period 1917-1939, is recorded as 1584 tons.

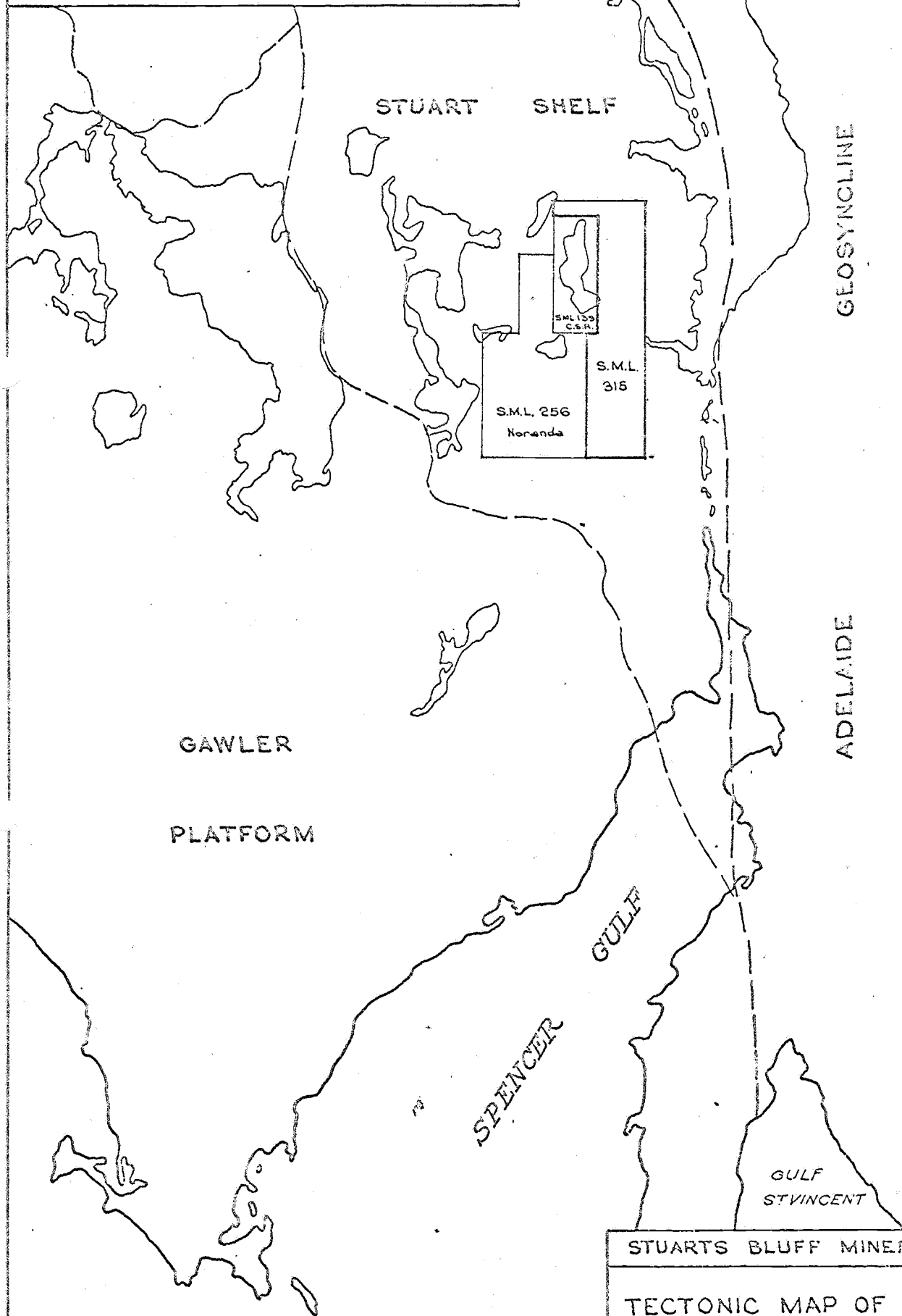
In reference to ore reserves the current estimate of 3,200,000 tons for the ore blocked out by C.S.R. and associates considerably upgrades proven copper reserves over past estimates. Unfortunately elsewhere little development prospecting appears to have been undertaken. The annual report on SML 55, July 1965, records potential reserves, based on geophysical and geochemical surveys, of 820,000 tons manganese for the main manganese deposits, and 200,000,000 tons of 0.5-1% copper for the northern copper deposits. It is not known to what extent exploratory drilling has been carried out to prove or disprove these estimates.

GEOLOGICAL DATA:

The deposition of the Pernatty Lagoon orebodies is a function of the stratigraphy and structure of the Pre-Cambrian sediments in which the orebodies occur. It follows that mineral exploration in adjacent areas requires a detailed knowledge of the stratigraphy and structure of these sediments. Before examining this aspect of exploration in SML 315 in detail, it is pertinent to review briefly the regional setting of the Pre-Cambrian sediments.



LOCALITY PLAN



STUARTS BLUFF MINERALS LTD.

TECTONIC MAP OF REGION
SURROUNDING S.M.L. 315

MILES 20 10 0 20 40 MILES

FIG. 2

STUART SHELF

Adelaide System:

Tent Hill Formation (Marinoan)
 Simmens Quartzite
 Corraberra Sandstone
 Tregalona Shale
 Whyalla Sandstone
 - - disconformity -

Woocalla Dolomite (Torrensian?)
 - disconformity -

Pandurra Formation (Torrensian?)
 - unconformity -

Roopena Volcanics (Willouran)

ADELAIDE GEOSYNCLINE

Adelaide System:

Wilpena Group (Marinoan)
 Umberatana Group (Marinoan-Sturtians)
 Burra Group (Torrensian)
 Callana Beds (Willouran)
 - unconformity -
 Callan Beds (Willouran)

including Roopena Volcanic Equivalents

- unconformity -

GAWLER PLATFORM

Carpentarian and Undifferentiated Lower Proterozoic rocks - folded metasediments and volcanics with extensive igneous intrusions.

Figure 2 depicts the geographic distribution of the three main tectonic units of the region, the Gawler Platform, Stuart Shelf and Adelaide Geosyncline, all composed of Pre-Cambrian (Proterozoic) rocks. The stratigraphic relationship of these rocks are shown in Table 1. Their outcrop distribution is illustrated in figure 3.

The Adelaide Geosyncline and Stuart Shelf are both composed of Upper Proterozoic rocks of the Adelaide System entirely sedimentary except for a basic volcanic sequence at the base. The two tectonic units are separated by a downwarped zone passing through Lake Torrens, Spencer Gulf and Gulf St. Vincent. The thick folded sequence of the Adelaide Geosyncline on passing westwards through the downwarped zone, gives way on the Stuart Shelf to a relatively thin, subhorizontal sequence of sandstone, shale and dolomite unconformably underlain by a basic volcanic sequence (Roopera Volcanics). To the south west the Gawler Platform forms a basement complex of intruded metasediments and volcanics on which the Stuart Shelf sequence rests with marked unconformity.

THE STUART SHELF:

The Pernatty Lagoon ore deposits and the adjacent area of SML 315 lie within this tectonic unit. The basal unit, the Roopera Volcanics, outcrops in one locality only, 25 miles south west of Port Augusta where it unconformably overlies the Gawler Platform complex and is unconformably by the Pandurra Formation. The remainder of the Stuart Shelf sequence is entirely composed of sediments, sub-horizontal, with dips of the order of 1° to 5° . The shallow regional dip of these sediments is to the north-east away from the Gawler Platform. The following is a brief description of the sequence in descending order:-

Tent Hill Formation - 4 conformable members:

Simmens Quartzite	white quartzite with shale casts.
Corraberra Sandstone	red-brown, flaggy micaceous sandstone.
Tregalona Shale	purple and brown siltstones and shale.
Whyalla Sandstone	white sandstone, conglomeratic at base.

Woocalla Dolomite - black manganiferous dolomite and black shale.

Pandurra Formation - red sandstone and conglomerate.

The three topmost members of the Tent Hill Formation have been eroded off much of southern area of the Stuart Shelf, their remnants forming the plateau of the Andamooka Range in the north and east, and the mesas west of Port Augusta and Spencer Gulf. The base of the escarpments of the plateau and mesas lies near the boundary between Tregalona Shale and Whyalla Sandstone.

South and west of the escarpment is the sand covered plain in which Whyalla Sandstone, Woocalla Dolomite and Pandurra Formation are poorly exposed. It is with these beds that the Pernatty Lagoon ore deposits are associated. The following is a reconstruction of the depositional history of these sediments:-

1. A shallow marine transgression across the Stuart Shelf accompanied by rapid deposition of the coarse-grained, cross-bedded sandstones and conglomerates of the Pandurra Formation.
2. Quiet deposition of the Woocalla Dolomite in shallow waters. The close association of manganese, iron and barium (and possibly copper) salts with dolomites and black shales suggests deposition in a partially closed basin or basins. The rims of the basin (or basins) consisted of Pandurra Formation sandstones as topographically high areas.
3. A further marine transgression accompanied by deposition of the basal conglomerate and the succeeding coarse-grained, cross-bedded sandstones of the Whyalla Sandstone. The latter completely covered the Woocalla Dolomite and transgressed onto the eroded surface of the Pandurra Formation.

The Pandurra Formation forms a prominent topographic high along the west side of Pernatty Lagoon. On the east slope of this high, the Whyalla Sandstone rests on the eroded surface of the Pandurra Formation. Further down slope, near the edge of the Lagoon, the Woocalla Dolomite separates the two sandstone units, the Pandurra Formation passing beneath the Woocalla Dolomite under the Lagoon, the Whyalla Sandstone lying above the Woocalla Dolomite.

In the northern copper deposits the copper ores are concentrated in the basal porous sandstone of Whyalla Sandstone, both in the lower slope where the sandstone rests on Woocalla Dolomite and on the upper slope where the sandstone lies on the eroded surface of the Pandura Formation. Minor concentrations of ore also occur in joints in the eroded dolomite near the Lagoon edge. Significantly ore concentrations do not appear in the Pandurra Formation.

In the southern copper workings the ore occurs as impregnations in black shale underlying the dolomite. Both dolomite and shale appear to pinch out against the sloping surface of the Pandurra Formation. No Whyalla Sandstone is present.

Similar thinning of the dolomite occurs in the vicinity of the manganese and barytes deposits. The manganese ores are concentrated by decomposition of manganiferous dolomite. The closely associated barytes is probably derived from the dolomite by a similar process of leaching and concentration.

The pinch out of the Woocalla Dolomite against the sloping surface of the Pandurra Formation probably represents an ancient Woocalla shoreline. Where Whyalla Sandstone transgresses both Woocalla Dolomite and Pandurra Formation, its basal porous sandstone apparently provides a conduit for the passage of copper bearing solutions and precipitation of copper ores from these solutions.

The conditions of stratigraphy and structure favouring deposition of copper, manganese and barytes ores in the exposed areas of Pernatty Lagoon might reasonably be expected to occur in the subsurface elsewhere in the region. The search for such deposits is the exploration objective in SML 315.

PREVIOUS EXPLORATION IN SML 315 AREA:

Aeromagnetic surveys, which included the area of SML 315, were flown in 1962 by the Bureau of Mineral Resources on behalf of the S.A. Mines Department. The results were subsequently interpreted by Austral Exploration Services on behalf of Noranda Aust. Pty. Ltd. A number of anomalies were defined in SML 315 but were of too great depth to be considered significant targets in the current exploration programme.

Geological and geochemical surveys were conducted over a 2700 square mile area (including the area of SML 315) in 1968 by D. Tonkin for Noranda. The main value of these surveys was the determination of the outcrop distribution of the Pandurra Formation and Whyalla sandstone and the establishment of the disconformity between these formations and the Woocalla Dolomite. Surface geochemical surveys are of little value in SML 315, much of the area is blanketed by sand and outcrops are poor. However the background values obtained over the whole region of study, for the Pandurra Formation, Woocalla Dolomite, and Whyalla Sandstone could prove useful in evaluating subsurface geochemical results.

THE EXPLORATION PROGRAMME IN SML 315:

The exploration objective in SML 315 is the discovery of ore deposits at shallow depth similar to those exposed on the west side of Pernatty Lagoon. As has been shown the determination of the stratigraphic and structural relationships of the Pandurra Formation, Woocalla Dolomite and Whyalla ^{SANDSTONE} in the area is an essential part of the exploration programme. The area is blanketed for the most part by recent sands and outcrops are limited. The exploration of the area from the outset therefore involves subsurface investigations by drilling or geophysical methods.

The exploration programme designed to cover the first year of operations is divided into two phases:

1. A refraction seismic survey to provide stratigraphic and structural data on the Whyalla Sandstone, Woocalla Dolomite and Pandurra Formation as a guide to phase 2.
2. A combined operation of shallow drilling and geochemical assay of drill cuttings as a guide to detailed drilling over selected localities.

Arrangements were made with the South Australian Department of Mines to undertake both phases of the programme with Departmental equipment and crews.

REFRACTION SEISMIC SURVEY:

The survey was initiated on 4th August and completed on 23rd August. A Texas Instruments 7000B 24 channel recorder with attendant equipment and crew was used to shoot 35 spreads on an approximate 4 square mile grid over an area of some 700 square miles. Experimental spreads shot over outcrop of Whyalla Sandstone, Woocalla Dolomite and Pandurra Formation yielded the following velocity data:

Whyalla Sandstone	5,000 - 7,000 f.p.s.
Woocalla Dolomite	8,500 - 12,000 f.p.s.
Pandurra Formation	12,500 - 16,000 f.p.s.

The exploration spreads yielded four velocity sets:

Set	
1	1,000 - 5,000 f.p.s.
2	5,000 - 9,000 f.p.s.
3	9,000 - 12,500 f.p.s.
4	12,500 - 16,000 f.p.s.

Isopach values, calculated for each of the four sets are given in Table 2 and figure 4.

Some divergence must be expected between the interpreted velocity and actual formation characteristics. However, if the derived velocity isopachs equate in general to Pandurra Formation, Woocalla Dolomite, Whyalla Sandstone and Cainozoic sediments respectively, then large areas exist where the same conditions of stratigraphy and structure are present as in the areas of known ore deposition at Pernatty Lagoon.

The absence of set 3 values in a number of areas suggests pinch-out of the Woocalla Dolomite. The isopach distribution illustrating this point is presented in figure 5. Onlap of Whyalla Sandstone onto Pandurra Formation is suggested by a comparison of isopach values shown on figure 4.

THE DRILLING PROGRAMME:

This operation has been hampered by mechanical problems associated with the drilling equipment. Insufficient air compression, leading to failure to remove drill cuttings and maintain penetration, resulted in frequent abandonment of drilling without fulfilling the desired objective. The collapse of a compressor piston, necessitating major workshop repairs, caused suspension of the operation on 27th November. The locations of the 16 holes spudded to date are shown on figure 6. The geochemical results are tabulated in the appendix.

The proposal for completion of this programme envisages 5,000' of drilling as the preliminary target, the drill holes averaging 100' in depth and penetrating Whyalla Sandstone and Woocalla Dolomite into the top of Pandurra Formation, with sampling of drill cuttings at five intervals and geochemical analysis with a field mounted atomic absorption spectrometer.

CONCLUSIONS AND RECOMMENDATIONS:

1. The stratigraphic and structural relationships of the Pernatty Lagoon ore deposits to the Pre-Cambrian sediments in which they occur favours sedimentary control of ore deposition.
2. Results of a refraction seismic survey undertaken in SML 315 justify the implementation of phase 2 of the first year operations involving 5,000' of shallow drilling and geochemical assay of drill cuttings. This latter operation has been suspended in its early stages following a major breakdown in drilling equipment.

It is recommended that the remainder of the drilling operation to be undertaken be concentrated in the area between Bookaloo and the southern end of Pernatty Lagoon.

D. ROSE.

TABLE 2 - SUMMARY OF SEISMIC RESULTS

485

Spread	Thickness		Set III	Depth to Set IV	Remarks
	Set I	Set II			
1	10	-	-	10	Pandurra o/c
2	15	-	100	115	Woocalla o/c
3	9	75	-	84	Whyalla o/c
4	10	-	45	55	Woocalla o/c
5	10	90	na	-	Whyalla o/c
6	32	33	-	65	
7	78	43	-	120	
8	120	110	-	230	
9	30	57	67	154	Bore 207
10	8	58	-	66	
11	20	167	-	187	
12	73	39	128	240	
13	21	-	78	99	
14	43	-	60	103	
15	20	81	-	101	
16	19	-	-	19	
17	22	100	162	284	
18	15	150	na	-	
19	22	41	134	197	
20	16	114	118	248	
21	24	-	-	24	
22	15	140	-	155	
23	17	106	120	243	Depth Probe
24	47	78	137	262	
25	26	-	-	26	
26	21	38	-	59	
27	23	64	102	189	
28	21	-	90	111	
29	19	45	-	64	
30	21	83	193	297	
31	19	28	67	114	
32	21	55	85	161	
33	19	93	-	112	
34	16	55	109	180	
35	20	24	200	244	Depth Probe Beda Bore